

PACL+: Online continual learning using proxy-anchor and contrastive loss with Gaussian replay for sensor-based human activity recognition

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ARTICLE INFO

Keywords:

Human activity recognition
Continual learning
Incremental learning
Contrastive learning
Catastrophic forgetting
Gaussian sampling

ABSTRACT

The increasing prevalence of wearable sensors and devices necessitates the development of Human Activity Recognition (HAR) systems that maintain accurate performance over time. Traditional HAR models, which rely on offline supervised training, struggle to adapt to the dynamic nature of real-world environments, leading to performance degradation due to catastrophic forgetting when new activities or users are introduced. In this paper, we propose a novel continual learning method (PACL+) that integrates Proxy Anchor loss, contrastive learning, and Gaussian replay to mitigate catastrophic forgetting in HAR systems and improve HAR performance. Unlike previous approaches, PACL+ effectively handles the introduction of both new activities and new users in incremental learning steps, addressing real-world challenges such as severe subject-wise class imbalance and user-dependent learning. To improve efficiency, we introduce Gaussian replay, a memory-efficient strategy that selects representative examples for rehearsal, further stabilizing the learning process. We evaluate PACL+ on three benchmark HAR datasets under realistic continual learning scenarios with varying sampling rates and diverse class distributions. Experimental results demonstrate that PACL+ significantly outperforms existing state-of-the-art methods, achieving higher accuracy and F1 scores while preserving performance on previously learned activities.

1. Introduction

With the ever-increasing demand and development of wearable sensor devices, developing systems that can sustain over time and provide accurate performance for Human Activity Recognition (HAR) is crucial. Accurate HAR enables real-time monitoring and analysis of physical movements, offering valuable applications in health monitoring (Sangeethalakshmi, Preethi, Pavithra et al., 2023), fitness tracking (Margarito, Helaoui, Bianchi, Sartor, & Bonomi, 2015; Singh, Ray, Zhou, Suh, & Lukowicz, 2024; Zhou, Suh, Rey, Altamirano, & Lukowicz, 2022a), worker safety (Bello, Geißler, Suh, Zhou, & Lukowicz, 2025; Suh et al., 2023), and personalized healthcare (Bello, Marin, Suh, Zhou, & Lukowicz, 2023; Islam, Nooruddin, Karray, & Muhammad, 2023). HAR systems leverage machine learning algorithms to analyze sensor data and classify physical activities, enabling real-time monitoring and intelligent decision-making. However, for HAR models to be truly effective in

real-world deployments, they must continuously adapt to new activities, new users, and evolving user behaviors while preserving previously learned knowledge.

Despite significant advancements, most existing HAR models are designed and evaluated in static offline settings, assuming that the environment, users, and their activities remain unchanged over time. While this approach has established a strong methodological foundation for HAR, state-of-the-art (SOTA) methods still fall short of bridging the gap between controlled conditions and dynamic real-world environments. A critical challenge in developing adaptive HAR systems lies in the need for trained models to integrate new knowledge while preserving previously learned information, avoiding the issue of catastrophic forgetting. Neural network models often lose previously acquired knowledge when learning new information, and balancing the stability of retaining old classes with the plasticity of learning new ones is a significant hurdle in continual learning, which is crucial for a sustainable HAR system.

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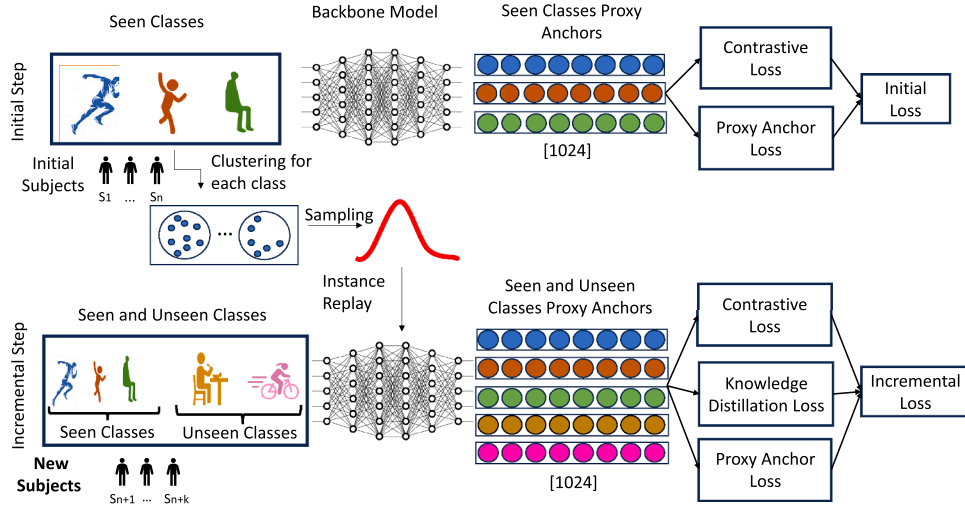


Fig. 1. Overview of the proposed framework. **Initial Step (Top):** The backbone model is trained on seen classes from the initial N subjects ($S_1, S_2, \dots, S_N$) using proxy anchor and contrastive loss. Class-wise clusters are then formed, and cluster statistics are saved so that samples can be drawn from each cluster, for instance, replay in the incremental step. **Incremental Step (Bottom):** The same backbone model now adapts to data from new subjects ($S_{N+1}, \dots, S_{N+K}$) that also contains new classes (generating new proxy anchors). Previous knowledge is kept using instance replay during training through proxy anchor loss, example replay, contrastive loss, and knowledge distillation.

Continual learning (CL) has gained considerable attention in machine learning, with applications ranging from object classification to high-end autonomous systems.

CL has been applied in various domains, such as learning new patient data for sepsis prediction in the medical field (Amrollahi, Shashikumar, Holder, & Nemati, 2022), navigation (Liu, Xiao, & Stone, 2021), and natural language processing (Jang, Kim, Choi, & Suh, 2021; Razdaibiedina et al., 2022). However, there are a limited number of existing works focusing on CL methods for HAR data, which is often complex and makes differentiating new activities challenging (Adaimi & Thomaz, 2022; Jha, Schiemer, Zambonelli, & Ye, 2021; Schiemer, Fang, Dobson, & Ye, 2023; Ye, Nakwijit, Schiemer, Jha, & Zambonelli, 2021). One such method, LAPNet-HAR (Adaimi & Thomaz, 2022), uses a task-free data-incremental paradigm with experience replay and continual prototype evolution to reduce catastrophic forgetting. Despite considering CL scenarios, this approach limits its scalability to large datasets due to the usage of a memory buffer and requires improvement in selecting representative samples to generalize embedding spaces for given classes. In addition, it overlooks performance degradation before and after new activities are added, a key challenge in CL.

Another widely used approach for continual learning is Elastic Weight Consolidation (EWC) (Kirkpatrick et al., 2017; Liu et al., 2018; Zhang et al., 2021), which mitigates catastrophic forgetting by constraining weight updates based on the importance of previously learned tasks. While EWC has shown promise in various applications, its reliance on estimating Fisher Information matrices (FIM) makes it

computationally expensive for real-time HAR applications, especially in resource-constrained environments (Dauncey, Holmes, Williams, & Falck, 2024). Moreover, EWC struggles in scenarios where the new activity distributions significantly differ from previously learned activities, leading to suboptimal adaptation and retention of knowledge in HAR settings.

Unlike general machine learning tasks such as computer vision or natural language processing, in practical scenarios, we often encounter the introduction of new activities over time, along with new subjects, which creates a data imbalance between previously seen and newly introduced unseen activities. This imbalance causes neural networks to forget patterns of previously seen activities, leading to overfitting on new unseen classes and subsequent degradation in performance on previously seen classes. Moreover, the inclusion of new subjects in the incremental step further intensifies the forgetting problem, as the neural network struggles to retain knowledge from earlier subjects.

To address these problems, we propose PACL+, as shown in Fig. 1. It is a novel method that combines Proxy Anchor loss (Kim et al., 2023; Kim, Kim, Cho, & Kwak, 2020) with contrastive learning on HAR data. This creates a better latent space representation that generalizes between subjects while also allowing the addition of new classes. Moreover, it treats the CL problem as a binary classifier, which reduces the computational demands compared to its predecessors, achieving remarkable success in computer vision and multimodal data (Radford et al., 2021). Typically, the end goal of HAR models is to be deployed in embedded devices (limited computation) to be pervasive.

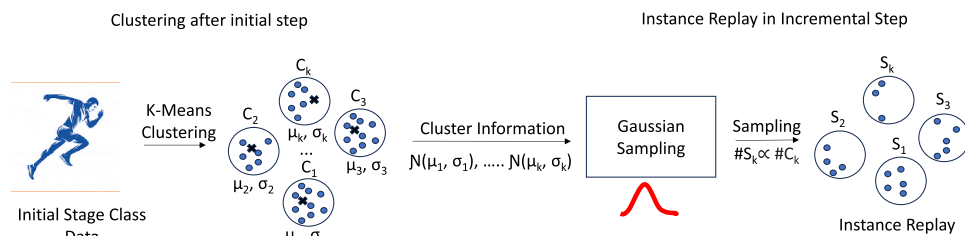


Fig. 2. Instance Gaussian replay workflow. For each class, multiple clusters are identified, and their statistical properties, such as mean, standard deviation, and cluster size, are computed. This information can then later be used to generate synthetic samples by sampling from the stored distributions. We promote a balanced distribution in the sampled data by generating samples for each cluster (S_k) proportionally to the number of samples in the original cluster (C_k).

Additionally, K-Means clustering is applied (see Fig. 2). This is followed by the Gaussian sampling method to obtain similar samples for the training in the initial step and perform example replay in the incremental step.

We evaluate the proposed method on three benchmark datasets: WISDM (Kwapisz, Weiss, & Moore, 2011), REALWORLD (Sztyley & Stuckenschmidt, 2016), and MHEALTH (Banos et al., 2014), and perform an extensive ablation study with realistic experimental settings for online continual learning scenarios. In addition, the proposed method was evaluated with two different network architectures: ResNet-18 and TinyHAR architecture (Zhou et al., 2022b), which is lightweight, efficient, and robust for human activity recognition tasks. The main contributions of this paper are summarized as follows.

- We present PACL+ (see Fig. 1), a novel method that combines proxy anchor loss with contrastive learning and Gaussian sampling approaches to improve the CL process in HAR. The hybrid approach effectively mitigates the issues of catastrophic forgetting through knowledge distillation and feature replay, enhancing knowledge retention across new and previously learned activities.
- Our method is especially suited for resource-constrained setups such as HAR as it has minimal storage needs. We show its resilience by evaluating it in realistic experimental settings with lightweight models for online CL scenarios by mixing seen and unseen classes in an incremental step.
- Our method performs efficient example replay by clustering data from previous steps using the K-means clustering algorithm and applying the Gaussian distribution sampling technique (Gaussian Replay).
- We validate our approach on three benchmark HAR datasets, WISDM, REALWORLD, and MHEALTH with the backbone neural network models including ResNet and TinyHAR, and compare them with state-of-the-art continual learning methods. These datasets vary in sampling rates and class distributions, providing a robust testbed for our proposed approach.

The rest of the paper is organized as follows. Section 2 introduces related works in incremental and continual learning in general and HAR. Section 3 provides a detailed description of the PACL+ framework, including knowledge distillation, exemplar replay, contrastive learning, and Gaussian sampling. Section 4 outlines our evaluation methodology, discusses our results, their implications, and current limitations, and Section 5 concludes this work with a discussion of future directions.

2. Related work

2.1. Incremental and continual learning in general

Continual learning (CL), also called lifelong learning, is the ability of an intelligence system to learn new tasks for which it was not originally trained (Liu, 2020). An intelligence system is expected to maintain or even improve performance on original tasks after learning new ones. For instance, humans can learn badminton, then tennis, and in the process, build endurance and learn techniques that enhance both sports and aid in learning a third. However, artificial intelligence (AI) systems often struggle to retain performance on original tasks when learning new ones, leading to catastrophic forgetting. Hence, CL is being extensively investigated to endow AI systems with this native capability in humans (Parisi, Kemker, Part, Kanan, & Wermter, 2019). Recently, the authors in Wang et al. (2023) presented VOYAGER. The solution interacts with GPT-4 via black-box queries to personify a lifelong learning agent in Minecraft that continuously explores the world, and acquires various skills without human intervention. The idea is to add new code tasks to a dictionary of skills and refine them based on the similarity between the old and new skills. VOYAGER is an example of continuous online meta-learning. This means that the AI system learns during

runtime instead of offline, where new tasks are statically predefined beforehand.

In general, CL methods are categorized as follows: regularization (Kirkpatrick et al., 2017), replay, representation (Rao et al., 2019), optimization, and architecture-based approaches, or a combination of them (De Lange et al., 2021; Wang, Zhang, Su, & Zhu, 2024).

The first version implies the use of penalization methods to prevent catastrophic forgetting. Weight and loss function regularization are commonly used to force the new model to retain memory information from the old model. This technique requires considering memory retention with reduced impact on plasticity to learn new tasks. To apply regularization-based CL efficiently is an open research question. Recently, in Shen, Ni, Xu, and Tang (2024), the spiking neural network (SNN) regularization method has been employed. Inspired by the selective sparse activation principle of context gating in biological systems, the authors use the SNN model with selective activation to achieve continual learning. The selective activation in spatial and temporal dimensions of SNN promotes the subpopulation of neuron activation to perform specific tasks. Therefore, different tasks will be routed through different populations of neurons in the network. CL with SNN is an example of regularization and architecture-based CL with the use of a selective activation function and task-specific redirection within the network (sub-network).

The replay approach involves retraining the new model with a few samples from a memory bank. The memory bank contains either real/generated data or feature samples of the original task. A balance between the representativeness of the memory sample and the practicality of the buffer size is required in this case. The main advantage of feature-based replay is that it preserves the memory of the original task without storing raw data, which helps reduce buffer size and maintain privacy. On the other hand, the mismatch between the original task features and dynamically changing features can cause several catastrophic forgetting. In Zhou et al. (2025), the authors present Manifold-Aligned Graph Regularization (MAGR). It reduces the misalignment between the old features and the new task features to ensure the consistency of the representation by employing a joint graph with quality scores. This idea combines regularization and repetition-based CL methods. Diffusion-driven data replay for Federated Class Continual Learning (FCCL) is introduced in Liang et al. (2025). Their solution leverages a pre-trained conditional diffusion model to generate samples by reverse-engineering each class to identify the corresponding input. This process is further refined through contrastive learning between real and generated data. As a result, the data generator becomes more stable, and the quality of the synthesized samples is enhanced. The replay/rehearsal CL method has shown potential for self-supervised continual learning. However, in Smith, Tian, Halbe, Hsu, and Kira (2023), the authors show that parameter rehearsal-free CL can also achieve state-of-the-art (SOTA) results and even outperform SOTA prompting for CL.

Architecture-based CL focuses on building task-specific parameters rather than a shared parameter space for the original and target tasks. It could also imply sub-networks inside the DNN to handle each task. In Li, Zhou, Wu, Socher, and Xiong (2019), the concept of learning to grow is introduced. The concept includes finding the optimal network structure for each of the sequential tasks, followed by parameter estimation. In this way, the optimized structure can alleviate catastrophic forgetting.

Overall, CL has strong potential for growth in research and its multiple applications. A notable application of CL is Human Activity Recognition (HAR) with time series data. Inertial measurement units (IMU) are a widely successful example of time series data used for HAR. However, this data type presents significant challenges due to both intra-class and inter-class variability, even for activities performed by the same individual together with activity duration variance. For example, a person may jump differently each time, making it nontrivial to determine this variation by analyzing IMU data and merging all instances into a single class description. In contrast, with images, it is intuitively easier to classify all jump styles simply as “jump” without much deliberation. These

complexities make continual learning particularly valuable for advancing HAR methodologies. In the next section, CL-specific SOTA for HAR is explored.

2.2. Continual learning in HAR

Most prior work on human activity recognition (HAR) has focused on offline supervised training. However, several incremental learning methods have been developed in recent years (Mazankiewicz, Böhm, & Bergés, 2020) to improve activity recognition performance (Ntalampiras & Roveri, 2016) and personalize HAR models across users by adapting to distribution shifts (Gjoreski & Roggen, 2017; Siirtola & Rönning, 2021). There is also growing interest in discovering and recognizing new activities for HAR. Popular techniques include clustering and one-class classifiers, where a new cluster/classifier is created for each new activity (Gjoreski & Roggen, 2017). However, simply adding new clusters/classifiers while fixing the data representation can lead to overlapping clusters, requiring full model retraining. Thus, evolving the HAR models with new data and activities is key for CL. CL allows a model to adapt to new information (unseen situations) and out-of-distribution (OOD) data examples without forgetting previously learned knowledge, overcoming catastrophic forgetting (Kirkpatrick et al., 2017). Unforeseen situations in HAR include new users or new tasks to identify, where adaptation to change is inevitable due to the dynamics of life. For example, in Chauhan, Kwon, Hui, and Mascolo (2020), CL is utilized to automatically adapt to new users, enhancing behavior-based user authentication.

The state of the art in CL for HAR identifies three main strategies: dynamic architectures, regularization, and rehearsal (Leite & Xiao, 2022; Schiemer et al., 2023).

Dynamic architectures adapt model structure over time by expanding capacity or adding expert modules. For instance, Net2Net (Chen, Goodfellow, & Shlens, 2015) and evolving CNNs (Ye & Callus, 2020) allow parameter growth as tasks accumulate. More recently, Liu, Bian, Zhou, and Lukowicz (2024) proposed iKAN, a continual learning framework using Kolmogorov-Arnold Networks, which excels at handling non-uniform inputs and achieves high performance on HAR datasets. Similarly, Rahman, Schiemer, Sanabria, and Ye (2025) introduced a dynamic mixture-of-experts model that creates new expert modules for each task and selects them adaptively, offering flexible scalability for embedded HAR systems. However, for wearable HAR, these dynamic architectures may negatively affect performance due to increased complexity, latency, and memory requirements.

Regularization methods add terms to the loss function to constrain weight updates to retain knowledge of old tasks while learning new ones. For example, Zhang et al. (2021) proposed a multi-modality incremental learning model, named HarMI, that adopted an attention mechanism to align sensor data, elastic weight consolidation, and canonical correlation analysis. Recently, Tang et al. (2024) explored the adaptation of continual self-supervised learning models, such as CaSSLe and Kaizen, for wearable-based HAR. Their study investigates the trade-offs between knowledge retention and learning new tasks, highlighting the importance of balancing objectives in continual learning.

Rehearsal methods involve replaying samples from previous tasks to train the network incrementally. Relevant data samples from seen classes are stored in memory or generated using generative adversarial networks (GANs) to produce intermediate CNN features for replay. Jha et al. (2021) evaluated various state-of-the-art CL techniques on HAR datasets, identifying the best-performing methods and their computational costs. Ye et al. (2021) applied GANs to generate sensor data for seen activity classes and mixed them with new class data to train an incremental classifier, addressing privacy concerns of storing real samples. However, these methods suffer from forgetting over time, and the quality of generated samples deteriorates, requiring larger batch sizes. LAPNet-HAR (Adaimi & Thomaz, 2022) constructed representative embeddings for a group of instances in a mini-batch, called prototypes.

The prototypes are continuously updated, and contrastive loss is used to enforce inter-class separation. LAPNet-HAR focuses on optimizing sample selection for replay-based techniques, leveraging gradient-based information to select samples that provide a diverse representation of the seen data. Similarly, Schiemer et al. (2023) proposed using both real samples and features for replay, resulting in an increase of 0.23 in the F1 score. Recent works have also explored dataset-level strategies to improve HAR robustness. Amrani, Micucci, Mobilio, and Napoletano (2025) presented a homogenization procedure to integrate diverse HAR datasets, enhancing model generalization and robustness. Their approach demonstrates the feasibility of data-centric strategies for robust HAR systems. Kanoga, Tsukiji, and Karakida (2025) evaluated 12 continual learning methods for user-centric class-incremental human activity recognition using IMU data, finding replay-based approaches most effective at mitigating forgetting with limited new data. Additionally, Zhuang et al. (2024) proposed Generalized Analytic Continual Learning (GACL), an exemplar-free, gradient-free method for GCIL that delivers a closed-form solution ensuring equivalence between incremental and joint training.

Nonetheless, existing methods still face challenges in scalability and effectively managing the balance between stability and plasticity in neural network models. CL techniques in HAR struggle with efficiently handling an ever-growing number of classes or activities. Computational cost is also a barrier, particularly when using complex memory or generative models.

In this paper, our proposed method (PACL+) addresses these limitations by combining a proxy anchor with contrastive learning on HAR data, providing a more robust approach to online CL in HAR. PACL+ uses a proxy anchor loss combined with contrastive learning to tackle scalability and stability challenges in HAR. Our method balances between maintaining old knowledge and adapting to new tasks, reducing catastrophic forgetting, and improving performance in both seen and unseen classes. We show how this approach can be effectively applied to HAR, offering a robust solution for continual learning in dynamic environments.

3. Methodology

Our method allows for new classes to be added along with new subjects incrementally while avoiding catastrophic forgetting of previously learned classes on previous subjects. A crucial component that enables our method to scale in resource-constrained environments is the use of Proxy Anchors (PA). We train a network f_θ that maps examples to embeddings. Instead of directly computing class probabilities, proxy networks learn an embedding for exemplars and represent each class as a point (proxy) in the embedding space. More formally, HAR samples x are tensors ($C \times T$) where C is the number of sensor channels and T is the size of the sliding window and we train a network $f_\theta : \mathbb{R}^{C \times T} \rightarrow \mathbb{R}^\epsilon$ where ϵ is the embedding size. Each class c is represented by a point $p_c \in \mathbb{R}^\epsilon$, called its proxy anchor. Proxies represent the center of each class and serve as anchors for optimizing the embeddings of all samples within the same class. Predictions for instances x are made by selecting its closest proxy anchor in the embedding space. More formally, this is expressed as $\arg \max_c s(f_\theta(x), p_c)$, where p_c is the anchor for class c and s is the cosine similarity $s(v_1, v_2) = \frac{v_1 \cdot v_2}{\|v_1\| \|v_2\|}$. This approach has the advantage of only requiring the storage of new anchors when adding new classes, rather than growing the underlying network structure. On the other hand, avoiding catastrophic forgetting of old proxy anchors when adding new ones can be a significant challenge.

In this Section, we present our method, which aims to handle catastrophic forgetting for HAR. The main steps, as illustrated in Fig. 1, are as follows: First, we learn a representation and proxies for the initial classes. From this step, we keep minimal information (only the old proxies and their feature distribution). Then, in the second step, where we have new data with both seen and unseen classes from different subjects, our approach can learn new proxies for unseen classes and update old

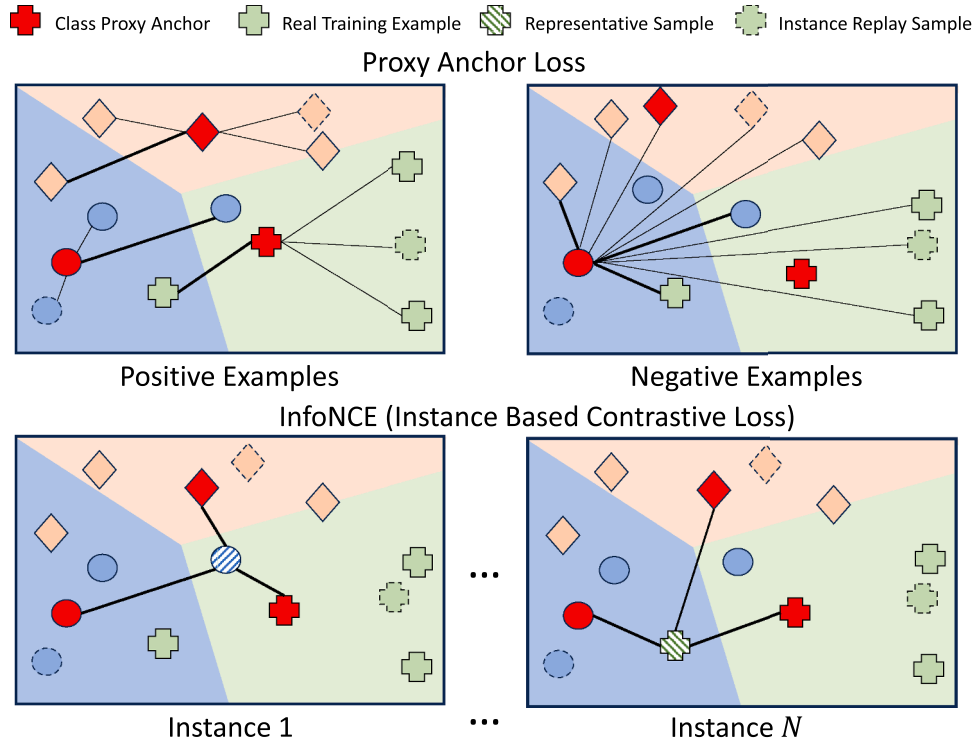


Fig. 3. Visual representation of our Proxy-Anchor loss (Top) and Contrastive Loss: “InfoNCE” (Bottom). Each proxy is depicted in red, while three distinct objects with varying colors and shapes are utilized to differentiate between classes. The connections established by the losses are visualized through edges (connection lines), with thicker edges denoting larger gradients.

proxies while avoiding catastrophic forgetting through Gaussian replay and knowledge distillation.

3.1. Initial step

In the initial step, we obtain proxies using both Proxy-Anchor (PA) loss and contrastive learning. PA loss, a novel approach in deep metric learning, addresses the limitations of existing proxy-based and pair-based methods, particularly their inefficiency in scaling with large datasets (Kim et al., 2020). The PA loss L_{pa} can be expressed as follows.

$$L_{pa}(\tilde{Z}) = \frac{1}{|P^+|} \sum_{p \in P^+} \log \left(1 + \sum_{z \in Z_p^+} e^{-\alpha(s(z,p)-\delta)} \right) + \frac{1}{|P|} \sum_{p \in P} \log \left(1 + \sum_{z \in Z_p^-} e^{\alpha(s(z,p)+\delta)} \right) \quad (1)$$

where P_+ represents the number of proxy anchors corresponding to the inputs in a batch, while $|P|$ represents the total number of classes. Additionally, Z_p^+ and Z_p^- denote the embedding output of the input data points belonging to and not to a specific proxy anchor p respectively. Here, $\delta > 0$ is a margin, and $\alpha > 0$ is a scaling factor. function $s(z, p)$ indicates again cosine similarity between sample embeddings $z = f_\theta(x)$ and class proxy anchor p . The first term in the loss function brings the data point closer to its respective proxy anchor whereas the second term simultaneously tries to separate them. Because of this consideration of data-to-data relations, PA loss provides richer supervisory signals to embedding networks during training. Fig. 3 shows how the proxy anchor method utilizes gradient scales to push or pull similar and dissimilar data points in the embedding space.

To further increase the mutual information between embeddings of points of a class and their anchors, we also use the information noise contrastive estimation loss (InfoNCE), a variant of Noise-Contrastive

Estimation (NCE) loss (van den, Li, & Vinyals, 2018). It is defined as

$$L_{InfoNCE}(\tilde{Z}) = -\log \left(\frac{e^{s(z_c, p_c)}}{e^{s(z_c, p_c)} + \sum_{i \neq c}^C e^{s(z_c, p_i)}} \right) \quad (2)$$

where z_c represents the embedding representation of a given datapoint of class c in the batch and p_c is the proxy anchor for class c (out of the C existing ones). As illustrated in Fig. 3, this loss focuses on the relative cosine similarity between each point and its correct Proxy Anchor with others, being thus an instance-based complement to the L_{pa} loss.

Thus, our total loss for the initial step is simply expressed as:

$$L_{Initial} = L_{pa} + L_{InfoNCE} \quad (3)$$

as we did not in practice find any need for weighting the contribution of each loss. Another advantage of our setup is that, since both losses rely on similarities within the batch, those can be computed only once and then reused in each loss.

3.2. Incremental step

In our incremental step, we encounter both new and old classes from completely different subjects. To mitigate the sudden shift in model parameters and weights caused by the introduction of new data with new classes and subjects, we employ knowledge distillation loss. Simultaneously, to prevent the model from forgetting the seen classes from previously encountered subjects, we sample data from the training set used in the initial step. This is done by clustering the class data using K-means clustering and then passing the cluster information, such as the mean and standard deviation, to sampling methods like Gaussian and KDE.

3.2.1. Knowledge distillation loss

Knowledge Distillation was initially utilized to replicate the knowledge and behavior of a large model (called the “teacher” model) to a simpler model (called the “student”) (Hinton, 2015). In the context of

incremental learning, it helps control the drastic changes in the output embedding of the seen classes. The mathematical formula is shown below in Eq. 4

$$L_{kd}(p_0, p) = \mathbb{E}_{\text{class } c \in p_0} \|p_0^c - p^c\|^2 \quad (4)$$

where we minimize the Euclidean distance between updated prototypes and their old versions. This technique collectively enhances the model's ability to generalize across both seen and unseen classes while minimizing forgetting.

3.2.2. Gaussian replay loss

Gaussian Replay enables the sampling of examples that are representative of those the model has encountered in previous steps. This is achieved by selecting a representative subset of examples from each cluster within a class. Specifically, for each class from the previous training set, we perform K-means clustering to divide the data into 10 clusters. Key statistical properties of these clusters, such as the mean, standard deviation, and cluster support are then extracted. Leveraging this information to capture the underlying cluster distribution enables the generation of synthetic while minimizing storage requirements. Our sampling procedure (depicted in Fig. 2) has synthetic samples are drawn from the predicted cluster distributions with the number of examples sampled from each cluster proportional to its support in the training dataset. This approach ensures that the sampled examples offer a balanced representation of the class, capturing its diversity and improving coverage. More formally, generating samples using our proposed Gaussian replay method follows Algorithm 1.

Algorithm 1 Full procedure for our Gaussian replay sampling.

Inputs:

Training data for the initial step

Number of samples to generate (*sample_budget*)

Output:

The generated *sample_budget* synthetic examples

Initial Step:

$n_{\text{classes}} \leftarrow 0$

for each class in initial step **do**

$n_{\text{classes}} \leftarrow n_{\text{classes}} + 1$

Perform k-means clustering on class instances

for each cluster of class **do**

Save mean (μ), standard deviation (σ), and support (N)

end for

end for

Incremental Step:

$X_s \leftarrow []$

$n_{\text{samples_per_class}} \leftarrow \frac{\text{sample_budget}}{n_{\text{classes}}}$

for each class c **do**

for each cluster of c with (μ_c, σ_c, N_c) **do**

$n_{\text{samples}} = n_{\text{samples_per_class}} \times \frac{N_c}{\sum_j N_j}$

Sample $n_{\text{samples}} \sim \mathcal{N}(\mu_c, \sigma_c)$

Add samples to X_s

end for

end for

return X_s

The sampled points are subsequently fed into the Proxy Anchor loss function L_{pa} , making our Proxy Anchor loss with example replay L_{ex} as described in Eq. 5.

$$L_{ex}(\tilde{Z}) = \frac{1}{2}(L_{pa}(\tilde{Z}) + L_{pa}(\tilde{Z}_S)) \quad (5)$$

where $\tilde{Z}_S$ corresponds to the embeddings of the Gaussian replay sampled data and $\tilde{Z}$ to the real training data. The sampled data is also

included in Eq. 2 and so we also have

$$L_{InfoNCE}(\tilde{Z}) = \frac{1}{2}(L_{InfoNCE}(\tilde{Z}) + L_{InfoNCE}(\tilde{Z}_S)) \quad (6)$$

This strategy not only aids the model in retaining knowledge of previously seen samples, mitigating catastrophic forgetting but also ensures a smooth integration of new data with prior knowledge. By anchoring new examples to well-represented clusters, the model achieves a more stable and generalized learning trajectory.

3.2.3. Total incremental loss

Adding $L_{InfoNCE}$ loss used in the initial step to Knowledge Distillation Loss and Example Replay loss, our total loss for the incremental step is

$$L_{Incremental} = w_{nce} L_{InfoNCE} + w_{ex} L_{ex} + w_{kd} L_{kd} \quad (7)$$

where w_{nce} , w_{ex} , and w_{kd} are weight values to balance losses between retaining old knowledge and improving the existing network.

4. Experimental results

4.1. Datasets and experimental design for realistic continual learning scenarios

We conducted our experiments on three widely used HAR benchmarks, as detailed in Table 1. Our experimental setup distinguishes between seen classes, which are present in both the initial and incremental steps, and unseen classes, which only appear in the incremental step for continual learning scenarios. To better reflect real-world conditions, we ensure that subjects in the initial step are entirely different from those in the incremental step. This separation prevents subject overlap and enforces a more challenging yet realistic learning scenario. The subject distribution across datasets is structured to highlight the challenge of adapting to new users with additional class data while retaining knowledge from previous ones. We executed the entire pipeline five times, varying the selection of subjects for both the initial and incremental steps in each run. Once all runs are complete, we calculate and report each evaluation metric's mean and standard deviation (e.g., Accuracy, F1 scores, Forgetting Accuracy) to provide a robust assessment of performance variability. This approach helps ensure that the results are not biased by any particular subject grouping and gives a clearer picture of how well the model generalizes across different subject distributions.

For consistency, we focus solely on accelerometer data while considering variations in sensor placement and class distributions across datasets:

- **WISDM (Wireless Sensor Data Mining)** (Kwapisz et al., 2011): contains sensor data from smartphones carried by 51 subjects while performing 18 different. The data is recorded at a sampling rate of 20 Hz, capturing accelerometer and gyroscope readings. We used brushing teeth, typing, walking, and writing as unseen classes in the incremental step, with the remaining 14 classes as seen classes in the initial and incremental steps.
- **REALWORLD** (Szttyler & Stuckenschmidt, 2016): includes sensor data from 15 subjects doing some of their daily activities. Sensors used several accelerometers, gyroscopes, and magnetometers attached to certain locations of the body, including the chest, forearm, head, shin, thigh, and upper arm, with a sampling rate of 50 Hz. The dataset contains 8 types of activities: walking, running, sitting, standing, lying, jumping, and climbing. We use sitting, lying, jumping, climbing up, and climbing down as the seen classes in the initial step. In the incremental step, we add the standing and walking classes as unseen classes.
- **MHEALTH** (Banos et al., 2014): is valuable for developing and evaluating human activity recognition systems. It includes sensor data from 10 subjects performing 12 activities, such as walking, running, jumping, sitting, standing, and cycling. The data is recorded at 50 Hz

Table 1

Descriptions of the benchmark datasets used in the experiments and the continual learning scenarios. WISDM (Kwapisz et al., 2011), REALWORLD (Szttyler & Stuckenschmidt, 2016), and MHEALTH (Banos et al., 2014).

Dataset	Sampling rate	Channels	#Subjects	#Classes	#Classes seen	#Unseen classes	#Initial subjects	#Incremental subjects
WISDM	20HZ	3	51	18	14	4	26	25
REALWORLD	50HZ	3	15	8	6	2	8	7
MHEALTH	50HZ	9	10	12	6	6	5	5

using accelerometers, gyroscopes, and electrocardiograms placed on the chest, right wrist, and left ankle. we used the sensors placed on the chest, right wrist, and left ankle. Regarding labels, we have L2 to L6 as initial classes and L7 to L12 as incremental ones.

For all datasets, we split them into equal window sizes of 2 seconds, without modifying the original sampling rate.

We compared our method with two established continual learning approaches: LAPNET-HAR (Adaimi & Thomaz, 2022) and Elastic Weight Consolidation (EWC) (Liu et al., 2018). Both methods represent distinct paradigms in continual learning-LAPNet-HAR as a rehearsal-based strategy with prototype replay, and EWC as a regularization-based method. To ensure fairness and reproducibility, we used the official implementations from their publicly available GitHub repositories.

Moreover, we designed other baselines for comparison:

- **Offline batch learning:** This serves as an upper bound for our method, representing a scenario where the initial step is the same as in the proposed method, but the incremental step includes training data from both the initial and incremental steps. This setup allows the model to learn from the entire dataset even at the incremental step, without any constraints imposed by the continual learning process.
- **Online fine-tuning:** This baseline approach does not rely on the instance replay method or knowledge distillation techniques during the incremental step. Instead, it directly fine-tunes the model on new data, potentially leading to catastrophic forgetting without leveraging any strategies to retain previously learned knowledge.

While recent continual learning approaches for HAR have emerged, for example, iKAN (Liu et al., 2024), CaSSLe and Kaizen adaptations (Tang et al., 2024), GACL (Zhuang et al., 2024), they were not included in the empirical comparison for several reasons. Firstly, many recent works do not provide publicly available code or pretrained models, which prevents fair and reproducible comparison. Secondly, several studies focus on different CL paradigms such as domain-incremental learning, multi-modal fusion, or self-supervised pretraining, which are not directly comparable to our online class-incremental HAR setup. Lastly, some methods require significantly more resources (e.g., multiple modalities, larger architectures, or ensemble strategies), making them less suitable for our targeted lightweight and real-time HAR systems. Despite these differences, we have thoroughly reviewed and discussed these recent contributions in our related works (Section 2) to position PACL+ within the broader landscape of continual learning in HAR.

To assess the performance of our proposed method with other chosen comparison methods, we used three evaluation metrics: accuracy, macro F1 score, and forgetting metric. Accuracy and macro F1 score are standard metrics commonly used in various HAR studies. The forgetting metric, on the other hand, measures the difference in performance on seen class data between the initial step and the incremental step, providing a way to quantify how much the performance deteriorates during the continual learning process.

We performed our experiments using two different backbone architectures: ResNet-18 and a lightweight TinyHAR model. The ResNet-18 model follows a standard ResNet-based architecture, consisting of two ResNet blocks per layer, where each block includes a 1D convolution, followed by BatchNorm1D and ReLU activation. The network was structured with the first three layers bringing the network to 64, then 128,

and finally 256 channels. The fourth and final layer comprised 1024 output units. At this stage, the output was flattened and passed through a fully connected layer to produce a 1024-dimensional embedding. On the other hand, TinyHAR (Zhou et al., 2022b) is designed for high accuracy with minimal computational overhead, making it particularly suitable for wearable and edge devices. It focuses on balancing high accuracy with low computational and memory requirements, making it ideal for real-time applications. The architecture typically uses techniques like depthwise separable convolutions to reduce the number of parameters and computational load, enabling it to process sensor data efficiently. It also incorporates methods to capture the temporal and spatial patterns of activities, such as recurrent layers (e.g., LSTMs or GRUs) or attention mechanisms. With its lightweight design and energy-efficient processing, TinyHAR is well-suited for deployment in portable and battery-powered devices, ensuring reliable HAR performance without draining resources. The last layer of the model is tuned to provide an embedding dimension of 1024. By leveraging these two architectures, our evaluation demonstrates the feasibility and scalability of the proposed method across both standard deep learning models and lightweight, energy-efficient solutions for real-time, continual human activity recognition.

ResNet-18 consists of 2.69 million parameters, requiring 32.73 GFLOPs and 17.36 GMACs. In comparison, TinyHAR offers a significantly more compact architecture, with only 808.93 thousand parameters, 13.40 GFLOPs, and 5.12 GMACs. This substantial reduction in computational complexity makes TinyHAR particularly well-suited for real-time and resource-constrained applications.

To evaluate the performance of each method, we ran each model on each dataset for 5 iterations with different subjects, comparing the results in terms of mean accuracy and F1 score, along with their respective standard deviations. The values of δ and α in Eq. 1 were selected as 16 and 0.4, respectively. We performed a warm-up for both the ResNet and TinyHAR models for 5 epochs to stabilize the learning process, where only the last layer and proxy anchor embeddings were updated. The embedding size for proxy anchors was chosen as 1024, and the input batch size was set to 256. Both the initial and incremental steps for each method were executed for 100 epochs, with early stopping after 10 epochs to prevent overfitting. In the incremental step, we generate new samples every epoch using a sample budget corresponding to 10 % of the incremental dataset size. Those samples are then added to the real training data of the incremental method. This was done for computational efficiency, but it is also possible to sample during the training procedure at each batch, reducing storage requirements.

The source code used in our experiments is available at: <https://github.com/Dhruvadyamittal/PACL>. The entire framework is implemented in Python using the PyTorch deep learning library, with additional support from Scikit-learn for model evaluation and performance metrics. All experiments were conducted on a workstation equipped with an NVIDIA RTX 8000 GPU, leveraging its high computational capacity to ensure efficient training and evaluation across multiple continual learning scenarios. The repository includes all scripts for model training, testing, and ablation studies, along with configuration files and detailed instructions to facilitate reproducibility and further research.

As discussed in Section 3, we employ three distinct loss functions: InfoNCE, Example Replay Loss, and Knowledge Distillation Loss, each scaled by different weight values. These weights are initially determined by computing the validation loss for each loss function before training, serving as an upper bound for the validation loss. After each iteration,

we dynamically adjust these weights by scaling them based on the ratio of the current validation loss to the current training loss, relative to the initial validation loss. This adaptive weighting ensures a balanced contribution of each loss function throughout the training process. The detailed algorithm for the procedure can be seen in [Algorithm 2](#). Changing the weight for the losses based on the validation loss is similar to scheduling procedures that scale the learning rate based on said validation loss.

Algorithm 2 Adaptive loss weighting strategy.

- 1: **Input:** Training dataset D_{train} , Validation dataset D_{val} , Loss functions $\mathcal{L}_{InfoNCE}$, $\mathcal{L}_{ER}$, $\mathcal{L}_{KD}$, Number of iterations T
 - 2: **Initialize:** Compute initial validation losses:
$$L_{InfoNCE}^{val}, L_{ER}^{val}, L_{KD}^{val} \leftarrow \text{Evaluate on } D_{val}$$
 - 3: Set initial loss weights: $L_{InfoNCE}^{val_init}$, $L_{ER}^{val_init}$, $L_{KD}^{val_init}$

$$L_{InfoNCE}^{val} \leftarrow L_{InfoNCE}^{val_init} \quad L_{ER}^{val} \leftarrow L_{ER}^{val_init} \quad L_{KD}^{val} \leftarrow L_{KD}^{val_init}$$
 - 4: **for** $t = 1$ to T **do**
 - 5: **for** $batch = 1$ to N **do**
 - 6: Compute current training loss $\mathcal{L}_{InfoNCE}^{train}$, $\mathcal{L}_{ER}^{train}$ and $\mathcal{L}_{KD}^{train}$ on the current batch
 - 7: Update loss weights:
$$w_{InfoNCE} \leftarrow \frac{L_{InfoNCE}^{val}}{L_{InfoNCE}^{train}} \times \frac{1}{L_{InfoNCE}^{val_init}}$$

$$w_{ER} \leftarrow \frac{L_{ER}^{val}}{L_{ER}^{train}} \times \frac{1}{L_{ER}^{val_init}}$$

$$w_{KD} \leftarrow \frac{L_{KD}^{val}}{L_{KD}^{train}} \times \frac{1}{L_{KD}^{val_init}}$$
 - 8: Train the model using weighted loss:
$$L_{train} = w_{InfoNCE} \mathcal{L}_{InfoNCE}^{train} + w_{ER} \mathcal{L}_{ER}^{train} + w_{KD} \mathcal{L}_{KD}^{train}$$
 - 9: Compute current validation loss $\mathcal{L}_{InfoNCE}^{val}$, $\mathcal{L}_{ER}^{val}$ and $\mathcal{L}_{KD}^{val}$, on D_{val} :
 - 10: **end for**
 - 11: **end for**
 - 12: **Output:** Trained model with adaptive loss weighting
-

4.2. Comparison results

In this subsection, we evaluate our proposed method PACL+, against baseline models and SOTA approaches, including LAPNET-HAR ([Adaimi & Thomaz, 2022](#)) and EWC ([Liu et al., 2018](#)), using ResNet-18 and TinyHAR as backbone models. To comprehensively assess its performance in mitigating catastrophic forgetting and improving overall recognition, we conduct two key evaluations. First, we measure the model's ability to retain knowledge of previously learned activities by analyzing the accuracy, F1 score, and forgetting metrics on the initial subjects during incremental learning. These evaluations, presented in [Tables 2 and 3](#), provide insights into the retention of learned representations across the incremental steps.

The results demonstrate that PACL+ consistently outperforms all baseline and SOTA methods, not only achieving superior performance in the initial step but also effectively preserving knowledge of previously seen classes after incremental learning. Compared to existing methods, PACL+ exhibits significantly lower forgetting, as reflected in its higher accuracy and F1 scores after new activities and subjects are introduced. Notably, our method achieves an improvement of 4–5% in accuracy in the incremental step, underscoring its robustness in maintaining learned representations over time while mitigating catastrophic forgetting.

Furthermore, as illustrated in [Fig. 4 and 5](#), we provide a visual comparison of accuracy trends after the initial and incremental steps. The offline baseline model exhibits the smallest accuracy decline, as expected, since it does not operate in an online continual learning setting. However, among online learning approaches, LAPNET-HAR ([Adaimi & Thomaz, 2022](#)) and online fine-tuning suffer from the most significant accuracy degradation, highlighting their struggles with catastrophic forgetting. In contrast, PACL+ consistently achieves the best performance, as indicated by its lower forgetting values and reduced standard deviation, showcasing its stability and robustness across different HAR scenarios. These results validate the effectiveness of our approach in overcoming the limitations of existing continual learning methods for sensor-based HAR.

To further assess the effectiveness of our proposed method, we evaluate its overall performance in the incremental step, considering both seen and unseen classes from entirely new subjects. The results, presented in [Tables 4 and 5](#) for the TinyHAR and ResNet-18 backbone models, respectively, report accuracy and macro F1 scores separately for seen and unseen classes, followed by the overall performance metrics. Across all datasets and in most cases, the proposed PACL+ consistently outperforms the baseline and the comparison methods, demonstrating its strong generalization ability even when new classes and subjects are introduced.

A holistic analysis of the results from [Tables 2–5](#) reveals that PACL+ excels not only in retaining knowledge of previously seen activities but also in achieving superior recognition of newly introduced classes performed by different subjects. This effectiveness stems from the integration of Proxy Anchor Loss, knowledge distillation, instance replay, and InfoNCE loss, which together facilitate stable and efficient continual learning while mitigating catastrophic forgetting.

Despite being designed for online continual learning in sensor-based HAR, LAPNet-HAR ([Adaimi & Thomaz, 2022](#)) exhibits performance degradation in our experimental setting due to its handling of prototype representations in a time-series context. LAPNet-HAR employs Prototypical Networks to address continual learning challenges in sensor-based Human Activity Recognition (HAR). To mitigate catastrophic forgetting, it incorporates example replay with a fixed buffer size. The prototype for each class, representing its embedding space, is computed as the average of all instance embeddings for that class using:

$$p^c = \frac{1}{|C_i|} \sum_{x_j \in C_i} f_\theta(x_j) \quad (8)$$

where C are classes, x are instances and f_θ is the embedding network.

However, this prototype averaging approach assumes a uniform distribution of samples, which does not align well with real-world sensor data and often exhibits high variability in temporal sampling, imbalanced distributions, and dynamic signal patterns. Consequently, LAPNet-HAR struggles to maintain class-specific temporal nuances, leading to suboptimal adaptation in incremental learning scenarios. Their model is also trained using the standard cross-entropy loss function, which may be less adaptive than our approach, which uses cosine similarity.

In contrast, our method demonstrates significantly lower forgetting of previously learned classes, as indicated by the marginal decline in accuracy and F1 scores for seen classes between the initial and incremental steps. On the other hand, LAPNet-HAR variants and the Online Fine-Tuning baseline experience a more substantial drop in performance, reflecting higher levels of catastrophic forgetting. Furthermore, our approach proves to be more resilient in handling imbalanced class distributions and incremental updates, further reinforcing its robustness and adaptability in real-world HAR scenarios.

4.3. Ablation study

To evaluate the effectiveness of each module in the proposed PACL+ framework, we conduct ablation studies in multiple ways. [Tables 6 and 7](#)

Table 2

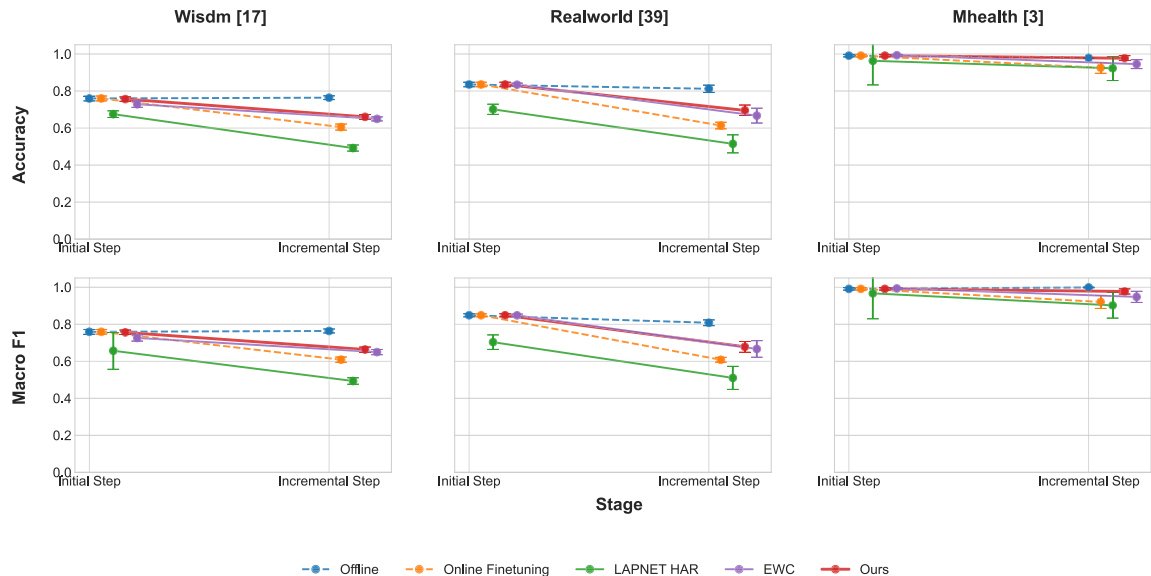
Comparison results of the proposed PACL+ evaluating the performance of initial subjects with baselines and other continual learning methods, including LAPNet-HAR (Adaimi & Thomaz, 2022) and EWC (Liu et al., 2018), using the TinyHAR backbone model.

Dataset	Method	Initial ACC $\uparrow$	Initial F1 $\uparrow$	Incremental ACC $\uparrow$	Incremental F1 $\uparrow$	Forgetting ACC $\downarrow$	Forgetting F1 $\downarrow$
WISDM	Offline	0.7599 ± 0.0132	0.7592 ± 0.0126	0.7641 ± 0.0101	0.7637 ± 0.0106	-0,0042	-0,0045
	Online Finetuning	0.7599 ± 0.0132	0.7592 ± 0.0126	0.6052 ± 0.0164	0.6087 ± 0.0143	01547	0,1505
	LAPNet-HAR	0.6751 ± 0.0180	0.6568 ± 0.01002	0.4919 ± 0.0165	0.4934 ± 0.0175	01832	01634
	EWC	0.7286 ± 0.0169	0.7261 ± 0.0170	0.6196 ± 0.0109	0.6196 ± 0.0140	01090	0,1065
	Ours (PACL+)	0.7599 ± 0.0132	0.7592 ± 0.0128	0.6605 ± 0.0131	0.6635 ± 0.0134	0.0994	0.0957
REALWORLD	Offline	0.8353 ± 0.0122	0.8485 ± 0.0081	0.8121 ± 0.0196	0.8081 ± 0.0156	00232	0,0404
	Online Finetuning	0.8353 ± 0.0122	0.8485 ± 0.0081	0.6135 ± 0.0178	0.6081 ± 0.0130	02218	0,2404
	LAPNet-HAR	0.7012 ± 0.0275	0.7033 ± 0.0393	0.5147 ± 0.0487	0.5102 ± 0.0623	01865	0,1931
	EWC	0.8342 ± 0.0084	0.8482 ± 0.0067	0.6671 ± 0.04	0.6666 ± 0.0449	01671	01816
	Ours (PACL+)	0.8353 ± 0.0122	0.8485 ± 0.0081	0.6953 ± 0.0290	0.6781 ± 0.0288	0.1400	0.1704
MHEALTH	Offline	0.9912 ± 0.0067	0.9912 ± 0.0069	0.9989 ± 0.0017	0.9989 ± 0.0016	-0,0077	-0,0077
	Online Finetuning	0.9912 ± 0.0067	0.9912 ± 0.0069	0.9257 ± 0.0301	0.9205 ± 0.0351	00655	0,0707
	LAPNet-HAR	0.9632 ± 0.1301	0.9669 ± 0.1372	0.9229 ± 0.0661	0.9020 ± 0.0689	0,0403	0,0649
	EWC	0.9935 ± 0.0047	0.9935 ± 0.0047	0.9454 ± 0.0241	0.9479 ± 0.0300	00481	00456
	Ours (PACL+)	0.9912 ± 0.0067	0.9912 ± 0.0069	0.9770 ± 0.0141	0.9772 ± 0.0139	0.0142	0.0140

Table 3

Comparison results of the proposed PACL+ evaluating the performance of initial subjects with baselines and other continual learning methods, including LAPNet-HAR (Adaimi & Thomaz, 2022) and EWC (Liu et al., 2018), using the ResNet-18 backbone model.

Dataset	Method	Initial ACC $\uparrow$	Initial F1 $\uparrow$	Incremental ACC $\uparrow$	Incremental F1 $\uparrow$	Forgetting ACC $\downarrow$	Forgetting F1 $\downarrow$
WISDM	Offline	0.7547 ± 0.0119	0.7537 ± 0.0109	0.7563 ± 0.0102	0.7553 ± 0.0095	-0,0016	-0,0016
	Online Finetuning	0.7547 ± 0.0119	0.7537 ± 0.0109	0.5895 ± 0.0195	0.5849 ± 0.0202	0.1652	0.1688
	LAPNet-HAR	0.6861 ± 0.0082	0.6866 ± 0.0077	0.4961 ± 0.0113	0.4941 ± 0.0126	0.1900	0.1925
	EWC	0.7461 ± 0.0082	0.7466 ± 0.0077	0.6327 ± 0.0128	0.6341 ± 0.0139	0.1134	0.1125
	Ours (PACL+)	0.7547 ± 0.0119	0.7537 ± 0.0109	0.6571 ± 0.0208	0.6814 ± 0.0221	0.0976	0.0723
REALWORLD	Offline	0.8461 ± 0.0182	0.8582 ± 0.0166	0.8270 ± 0.0199	0.8421 ± 0.0167	0,0191	0,0161
	Online Finetuning	0.8461 ± 0.0182	0.8582 ± 0.0166	0.6154 ± 0.0180	0.6385 ± 0.0206	0.2307	0.2197
	LAPNet-HAR	0.7812 ± 0.0275	0.7921 ± 0.0393	0.6211 ± 0.0487	0.6302 ± 0.0623	0.1601	0.1619
	EWC	0.8410 ± 0.0078	0.8550 ± 0.0078	0.6554 ± 0.0245	0.6548 ± 0.0196	0.1856	0.2002
	Ours (PACL+)	0.8461 ± 0.0182	0.8582 ± 0.0166	0.7095 ± 0.0338	0.7192 ± 0.0316	0.1366	0.1390
MHEALTH	Offline	0.9918 ± 0.005	0.9915 ± 0.0052	1.0000 ± 0.0000	1.0000 ± 0.0000	-0,0082	-0,0085
	Online Finetuning	0.9918 ± 0.0050	0.9915 ± 0.0052	0.8952 ± 0.0458	0.8816 ± 0.0579	00966	01099
	LAPNet-HAR	0.9541 ± 0.0397	0.9527 ± 0.0404	0.9427 ± 0.0322	0.9429 ± 0.0312	00114	0,0098
	EWC	0.9983 ± 0.0023	0.9983 ± 0.0023	0.9304 ± 0.0632	0.9382 ± 0.0749	0.0679	0.0601
	Ours (PACL+)	0.9918 ± 0.0050	0.9915 ± 0.0052	0.9899 ± 0.0007	0.9934 ± 0.0007	0.0019	-0.0019

**Fig. 4.** Initial and incremental stage performance on TinyHAR backbone model.

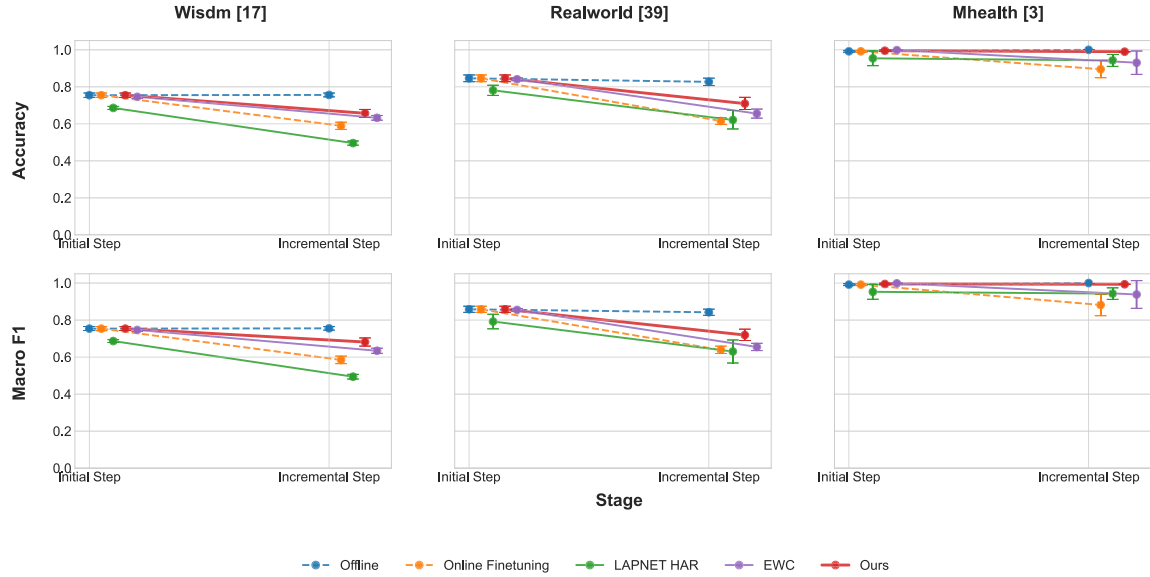


Fig. 5. Initial and incremental stage performance on Resnet18 Backbone model.

Table 4

Comparison results of the proposed PACL+ evaluating the performance of the incremental step in terms of accuracy and F1 score with baselines and other continual learning methods using the TinyHAR backbone model.

Dataset	Method	ACC (Seen)↑	F1 (Seen)↑	ACC (Unseen) ↑	F1 (Unseen)↑	Overall ACC↑	Overall F1↑
WISDM	Offline	0.7557 ± 0.0117	0.7517 ± 0.0099	0.8752 ± 0.0156	0.8879 ± 0.0129	0.7823 ± 0.0102	0.782 ± 0.0093
	Online Finetuning	0.7618 ± 0.0138	0.7645 ± 0.0119	0.8913 ± 0.0110	0.8777 ± 0.0165	0.7906 ± 0.0121	0.7896 ± 0.0113
	LAPNet-HAR	0.6251 ± 0.0361	0.6219 ± 0.0277	0.8200 ± 0.0039	0.8101 ± 0.0042	0.6683 ± 0.0121	0.6780 ± 0.0721
	EWC	0.7449 ± 0.0101	0.7437 ± 0.0082	0.8858 ± 0.0103	0.8826 ± 0.0125	0.7763 ± 0.0087	0.7745 ± 0.0080
	Ours (PACL+)	0.7777 ± 0.0161	0.7584 ± 0.0152	0.9094 ± 0.0093	0.8848 ± 0.0115	0.8170 ± 0.0140	0.8100 ± 0.0132
REALWORLD	Offline	0.8057 ± 0.0179	0.8199 ± 0.0173	0.8227 ± 0.0104	0.8236 ± 0.0147	0.8107 ± 0.0121	0.8208 ± 0.0101
	Online Finetuning	0.8066 ± 0.0167	0.8390 ± 0.0148	0.8515 ± 0.0152	0.7986 ± 0.0162	0.8198 ± 0.0131	0.8289 ± 0.0135
	LAPNet-HAR	0.6505 ± 0.06470	0.6761 ± 0.0924	0.6525 ± 0.1382	0.7171 ± 0.03914	0.6512 ± 0.0614	0.6435 ± 0.0262
	EWC	0.8121 ± 0.0143	0.8482 ± 0.0076	0.8659 ± 0.0146	0.8067 ± 0.0207	0.8280 ± 0.0103	0.8378 ± 0.0099
	Ours (PACL+)	0.8281 ± 0.0078	0.8316 ± 0.0061	0.8656 ± 0.0111	0.7931 ± 0.0163	0.8321 ± 0.0049	0.8409 ± 0.0046
MHEALTH	Offline	0.9989 ± 0.0010	0.9987 ± 0.0012	0.9971 ± 0.0042	0.9976 ± 0.0034	0.9981 ± 0.0017	0.9982 ± 0.0016
	Online Finetuning	0.9540 ± 0.0011	0.9986 ± 0.0012	0.9672 ± 0.0015	0.9983 ± 0.0012	0.9684 ± 0.0012	0.9984 ± 0.0011
	LAPNet-HAR	0.95515 ± 0.0906	0.9499 ± 0.0998	0.6053 ± 0.0571	0.5676 ± 0.0562	0.7833 ± 0.0697	0.7959 ± 0.0853
	EWC	1.0000 ± 0.0000	1.0000 ± 0.0000	0.9959 ± 0.0049	0.9967 ± 0.0041	0.9978 ± 0.0021	0.9979 ± 0.002
	Ours (PACL+)	1.0000 ± 0.0000	0.9992 ± 0.0011	0.9965 ± 0.0062	0.9961 ± 0.0059	0.9978 ± 0.0028	0.9976 ± 0.0032

Table 5

Comparison results of the proposed PACL+ evaluating the performance of the incremental step in terms of accuracy and F1 score with baselines and other continual learning methods using the ResNet-18 backbone model.

Dataset	Method	ACC (Seen)↑	F1 (Seen)↑	ACC (Unseen) ↑	F1 (Unseen)↑	Overall ACC↑	Overall F1↑
WISDM	Offline	0.7461 ± 0.0080	0.7432 ± 0.0070	0.8837 ± 0.0094	0.8909 ± 0.0118	0.7768 ± 0.0073	0.7760 ± 0.0059
	Online Finetuning	0.7463 ± 0.0172	0.7610 ± 0.0144	0.8924 ± 0.0146	0.8877 ± 0.0105	0.7890 ± 0.0135	0.7892 ± 0.0131
	LAPNet-HAR	0.6510 ± 0.0141	0.6580 ± 0.0153	0.8139 ± 0.0190	0.8160 ± 0.0152	0.7262 ± 0.0132	0.7162 ± 0.0510
	EWC	0.7395 ± 0.0129	0.7361 ± 0.0119	0.8821 ± 0.0205	0.8767 ± 0.0109	0.7713 ± 0.0131	0.7673 ± 0.0108
	Ours (PACL+)	0.7756 ± 0.0169	0.7748 ± 0.0155	0.8849 ± 0.0170	0.8996 ± 0.0143	0.8066 ± 0.0140	0.8068 ± 0.0133
REALWORLD	Offline	0.8175 ± 0.0108	0.8375 ± 0.0121	0.8458 ± 0.0246	0.8316 ± 0.0219	0.8259 ± 0.0111	0.8360 ± 0.0096
	Online Finetuning	0.8050 ± 0.0150	0.8429 ± 0.0078	0.8733 ± 0.0296	0.8066 ± 0.0208	0.8251 ± 0.0103	0.8338 ± 0.0090
	LAPNet-HAR	0.6702 ± 0.0612	0.6812 ± 0.032	0.6625 ± 0.1382	0.7111 ± 0.03914	0.6622 ± 0.0414	0.6535 ± 0.0212
	EWC	0.8136 ± 0.0130	0.8462 ± 0.0107	0.8513 ± 0.0211	0.8042 ± 0.0253	0.8146 ± 0.0144	0.8357 ± 0.0125
	Ours (PACL+)	0.8221 ± 0.0228	0.8269 ± 0.0128	0.8756 ± 0.0154	0.7972 ± 0.0195	0.8308 ± 0.0131	0.8195 ± 0.0125
MHEALTH	Offline	1.0000 ± 0.0000	1.0000 ± 0.0000	0.9984 ± 0.0025	0.9986 ± 0.0022	0.9993 ± 0.0012	0.9993 ± 0.0011
	Online Finetuning	1.0000 ± 0.0000	0.9999 ± 0.0003	0.9978 ± 0.0024	0.9982 ± 0.0021	0.9990 ± 0.0011	0.9990 ± 0.0010
	LAPNet-HAR	0.9478 ± 0.0214	0.9487 ± 0.0216	0.8097 ± 0.0525	0.8613 ± 0.0389	0.8828 ± 0.0260	0.8834 ± 0.0272
	EWC	1.0000 ± 0.0000	1.0000 ± 0.0003	0.9971 ± 0.0034	0.9975 ± 0.0036	0.9987 ± 0.0016	0.9986 ± 0.0018
	Ours (PACL+)	1.0000 ± 0.0000	1.0000 ± 0.0000	0.999 ± 0.0013	0.9992 ± 0.0011	0.9996 ± 0.0006	0.9996 ± 0.0006

Table 6

Ablation study for the impact of embedding dimension on the proposed method on initial subjects on both TinyHAR and Resnet18 architecture on REALWORLD (Szttyler & Stuckenschmidt, 2016) dataset.

Arc	Embedding Size	Initial ACC ↑	Initial F1 ↑	Incremental ACC ↑	Incremental F1 ↑	Forgetting ACC ↓	Forgetting F1 ↓
TINYHAR	256	0.8210 ± 0.0201	0.8312 ± 0.0092	0.6800 ± 0.0282	0.6591 ± 0.0300	0.1410	0.1721
	512	0.8259 ± 0.02010	0.8319 ± 0.0072	0.6802 ± 0.0219	0.6609 ± 0.0421	0.1457	0.1710
	1024	0.8353 ± 0.0122	0.8485 ± 0.0081	0.6953 ± 0.0290	0.6781 ± 0.0288	0.1400	0.1704
	2048	0.8320 ± 0.0131	0.8399 ± 0.0095	0.6810 ± 0.0231	0.6671 ± 0.0300	0.1510	0.1728
Resnet18	256	0.8300 ± 0.0191	0.8480 ± 0.0200	0.6891 ± 0.0491	0.7012 ± 0.041	0.1409	0.1468
	512	0.8310 ± 0.0151	0.8500 ± 0.0211	0.6901 ± 0.0313	0.7090 ± 0.0150	0.1409	0.1408
	1024	0.8461 ± 0.0182	0.8582 ± 0.0166	0.7095 ± 0.0338	0.7192 ± 0.0316	0.1366	0.1390
	2048	0.8400 ± 0.0199	0.8510 ± 0.0121	0.6912 ± 0.0331	0.7050 ± 0.0121	0.1488	0.1460

Table 7

Ablation study for the impact of embedding dimension on the proposed method on incremental subjects on both TinyHAR and Resnet18 architecture on REALWORLD (Szttyler & Stuckenschmidt, 2016) dataset.

Arc	Embedding Size	ACC (Seen)	F1 (Seen)	ACC (Unseen)	F1 (Unseen)	Overall ACC	Overall ACC
TINYHAR	256	0.8100 ± 0.0069	0.8210 ± 0.0067	0.8500 ± 0.0211	0.7801 ± 0.0199	0.8211 ± 0.0056	0.8309 ± 0.0056
	512	0.8200 ± 0.0065	0.8291 ± 0.0079	0.8571 ± 0.0291	0.7810 ± 0.0190	0.8300 ± 0.0091	0.8314 ± 0.0051
	1024	0.8281 ± 0.0078	0.8316 ± 0.0061	0.8656 ± 0.0111	0.7931 ± 0.0163	0.8321 ± 0.0049	0.8409 ± 0.0046
	2048	0.8210 ± 0.0090	0.8219 ± 0.0078	0.8510 ± 0.0016	0.7890 ± 0.0210	0.8291 ± 0.0056	0.8399 ± 0.0039
Resnet18	256	0.8191 ± 0.0321	0.8152 ± 0.0131	0.8759 ± 0.1151	0.7973 ± 0.0199	0.8293 ± 0.0214	0.8099 ± 0.0251
	512	0.8093 ± 0.0210	0.8091 ± 0.0310	0.8610 ± 0.1290	0.7851 ± 0.0210	0.8091 ± 0.0282	0.7999 ± 0.0263
	1024	0.8221 ± 0.0228	0.8269 ± 0.0128	0.8756 ± 0.0154	0.7972 ± 0.0195	0.8308 ± 0.0131	0.8195 ± 0.0125
	2048	0.8190 ± 0.0239	0.8176 ± 0.0213	0.8600 ± 0.1714	0.7810 ± 0.0516	0.8219 ± 0.0161	0.8091 ± 0.0314

present the ablation study justifying the selection of 1024 as the embedding size for all our experiments. We also conducted ablation studies on other hyperparameters, such as α and δ ; however, these parameters had a relatively minor impact on overall model performance. Therefore, we do not include their results in this manuscript. The first study examines the impact of contrastive learning using InfoNCE loss within our framework. Specifically, we compare Offline Batch Learning and Online Finetuning, both with and without InfoNCE loss, to analyze its contribution to performance. Table 8 presents the results for initial subjects after both the initial and incremental steps, highlighting the effect of contrastive loss on Offline and Online Finetuning to examine the degree of catastrophic forgetting across both backbone architectures. Additionally, Tables 9 and 10 examine the performance impact in the incremental step, further comparing models with and without contrastive loss on the TinyHAR and ResNet-18 architectures, respectively. The results indicate that incorporating InfoNCE contrastive loss leads to substantial improvements in overall accuracy and F1 score, while also ensuring greater stability, as evidenced by lower standard deviations. This demonstrates that contrastive learning effectively enhances feature discrimination and mitigates catastrophic forgetting, making it a crucial component of our proposed approach.

To further analyze the contribution of each module in our proposed PACL+ framework, we conducted a detailed ablation study. The following experimental settings were designed to assess the role of contrastive learning, replay mechanisms, and loss formulations in our framework:

- **No Contrastive:** This variant removes InfoNCE loss, relying solely on Proxy Anchor loss, example replay, and knowledge distillation. This setup isolates the impact of contrastive learning on performance.
- **Pairwise Contrastive:** Instead of InfoNCE loss, this version employs pairwise contrastive loss, as defined in Eq. 9, to evaluate its effectiveness compared to InfoNCE-based contrastive learning.
- **No Gaussian Replay:** This version of our proposed method omits the example replay mechanism, focusing solely on the Proxy Anchor loss and knowledge distillation to evaluate the impact of replay on continual learning performance.
- **PACL+:** The complete proposed method, incorporating Proxy Anchor loss, InfoNCE loss, knowledge distillation, and Gaussian replay,

representing the best-performing version of our framework.

$$\mathcal{L}_{\text{pairwise}} = \frac{1}{N} \sum_{i=1}^N \left[Y_{ij} \cdot d_{ij}^2 + (1 - Y_{ij}) \cdot \max(0, m - d_{ij})^2 \right] \quad (9)$$

where N is the total number of pairs, Y_{ij} denotes pair similarity, d_{ij} is the embedding distance, and m is the margin hyperparameter set to 1.0, as suggested in Adaimi and Thomaz (2022).

The results shown in Tables 11 and 12 show performance comparisons of methods selected for the ablation study on initial step test data before and after incremental step, whereas Tables 13 and 14 show the performance on the incremental step on TinyHAR and Resnet18 backbone architecture respectively.

The results in Tables 11 and 12 demonstrate that the full PACL+ model significantly outperforms its ablated versions, confirming the importance of each component in enhancing continual learning performance. Notably, Gaussian replay (No Gaussian Replay) leads to a marked drop in all metrics, accuracy, F1 scores, and forgetting metrics, highlighting the critical role of Gaussian replay sampling. Similarly, the removals of contrasting learning and InfoNCE loss result in higher forgetting, illustrating the necessity of InfoNCE loss.

Additionally, Tables 13 and 14 examine the model's performance on incremental step data, shedding light on how different loss components affect learning stability and adaptation. The findings indicate that the inclusion of Gaussian replay significantly helps to retain information from previous steps, leading to improved accuracy during the incremental step while keeping up with good classification scores for the incremental step data.

These findings are further supported by t-SNE visualizations (Devassy & George, 2020) in Fig. 6, which depicts the representation of our data and proxy anchors in the embedding space in the RealWorld dataset. These visualizations highlight significant changes in the embedding space from the initial to the incremental step, which contribute to catastrophic forgetting.

Moreover, to assess the impact of different sampling strategies on performance and computational efficiency, we performed an ablation study comparing Gaussian instance sampling and Kernel Density Estimation (KDE). Although KDE provides a more accurate approximation of the data distribution, it incurs higher computational memory

Table 8

Ablation study for the impact of InfoNCE loss evaluating the performance for initial subjects on Offline batch learning and online fine-tuning.

Arc	Dataset	Method	Initial ACC ↑	Initial F1 ↑	Incremental ACC ↑	Incremental F1 ↑	Forgetting ACC ↓	Forgetting F1 ↓
TINYHAR	WISDM	Offline (No NCE)	0.7336 ± 0.0134	0.7332 ± 0.0136	0.7304 ± 0.0130	0.7266 ± 0.0149	0.0032	0.0066
		Offline	0.7599 ± 0.0132	0.7592 ± 0.0126	0.7641 ± 0.0101	0.7637 ± 0.0106	-0.0042	-0.0045
		Online (No NCE)	0.7322 ± 0.0128	0.7308 ± 0.0158	0.5865 ± 0.0128	0.5882 ± 0.0116	0.1457	0.1426
		Online	0.7599 ± 0.0132	0.7592 ± 0.0126	0.6052 ± 0.0164	0.6087 ± 0.0143	0.1547	0.1505
	REALWORLD	Offline (No NCE)	0.8250 ± 0.0105	0.8398 ± 0.0077	0.8049 ± 0.0171	0.8004 ± 0.0119	0.0201	0.0394
		Offline	0.8353 ± 0.0122	0.8485 ± 0.0081	0.8121 ± 0.0196	0.8081 ± 0.0156	0.0232	0.0404
		Online (No NCE)	0.8250 ± 0.0105	0.8398 ± 0.0077	0.6065 ± 0.0342	0.6038 ± 0.0388	0.2285	0.2460
		Online	0.8353 ± 0.0122	0.8485 ± 0.0081	0.6135 ± 0.0178	0.6081 ± 0.0130	0.2218	0.2404
	MHEALTH	Offline (No NCE)	0.9746 ± 0.0027	0.9746 ± 0.0027	0.9592 ± 0.0011	0.5992 ± 0.0011	0.0154	0.0154
		Offline	0.9912 ± 0.0067	0.9912 ± 0.0069	0.9989 ± 0.0017	0.9989 ± 0.0016	-0.0077	-0.0077
		Online (No NCE)	0.9746 ± 0.0027	0.9746 ± 0.0027	0.9018 ± 0.0404	0.9011 ± 0.0518	0.0728	0.0735
		Online	0.9912 ± 0.0067	0.9912 ± 0.0069	0.9257 ± 0.0301	0.9205 ± 0.0351	0.0655	0.0707
Resnet18	WISDM	Offline (No NCE)	0.7461 ± 0.0082	0.7466 ± 0.0077	0.7293 ± 0.0064	0.7259 ± 0.0108	0.0168	0.0207
		Offline	0.7547 ± 0.0119	0.7537 ± 0.0109	0.7563 ± 0.0102	0.7553 ± 0.0095	-0.0016	-0.0016
		Online (No NCE)	0.7461 ± 0.0082	0.7466 ± 0.0077	0.5764 ± 0.0117	0.5702 ± 0.0119	0.1697	0.1764
		Online	0.7547 ± 0.0119	0.7537 ± 0.0109	0.5895 ± 0.0195	0.5849 ± 0.0202	0.1652	0.1688
	REALWORLD	Offline (No NCE)	0.8242 ± 0.0078	0.8350 ± 0.0078	0.7871 ± 0.0228	0.8197 ± 0.0194	0.0039	0.0153
		Offline	0.8461 ± 0.0182	0.8582 ± 0.0166	0.8270 ± 0.0199	0.8421 ± 0.0167	0.0190	0.0161
		Online (No NCE)	0.8242 ± 0.0078	0.8350 ± 0.0078	0.5748 ± 0.0267	0.6031 ± 0.0325	0.2462	0.2319
		Online	0.8461 ± 0.0182	0.8582 ± 0.0166	0.6154 ± 0.0180	0.6385 ± 0.0206	0.2307	0.2197
	MHEALTH	Offline (No NCE)	0.9869 ± 0.0029	0.9868 ± 0.0030	1.0000 ± 0.0000	1.0000 ± 0.0000	-0.0031	-0.0032
		Offline	0.9918 ± 0.005	0.9915 ± 0.0052	1.0000 ± 0.0000	1.0000 ± 0.0000	-0.0082	-0.0085
		Online (No NCE)	0.9869 ± 0.0029	0.9868 ± 0.0030	0.8764 ± 0.0389	0.8750 ± 0.0517	0.1105	0.1118
		Online	0.9918 ± 0.0050	0.9915 ± 0.0052	0.8952 ± 0.0458	0.8816 ± 0.0579	0.0966	0.1099

Table 9

Ablation study for the impact of InfoNCE loss evaluating the performance for the incremental step on Offline batch learning and online fine-tuning using the TinyHAR backbone model.

Dataset	Method	ACC (Seen)	F1 (Seen)	ACC (Unseen)	F1 (Unseen)	Overall ACC	Overall F1
WISDM	Offline (No NCE)	0.7293 ± 0.0122	0.7200 ± 0.0115	0.8470 ± 0.0141	0.8804 ± 0.0108	0.7555 ± 0.0116	0.7557 ± 0.0111
	Offline	0.7557 ± 0.0117	0.7517 ± 0.0099	0.8752 ± 0.0156	0.8879 ± 0.0129	0.7823 ± 0.0102	0.782 ± 0.0093
	Online (No NCE)	0.7520 ± 0.0155	0.7522 ± 0.0152	0.8817 ± 0.0115	0.8793 ± 0.0091	0.7731 ± 0.0141	0.7827 ± 0.0131
	Online	0.7618 ± 0.0138	0.7645 ± 0.0119	0.8913 ± 0.0110	0.8777 ± 0.0165	0.7906 ± 0.0121	0.7896 ± 0.0113
REALWORLD	Offline (No NCE)	0.7960 ± 0.0148	0.8053 ± 0.0162	0.8100 ± 0.0241	0.8052 ± 0.0190	0.7931 ± 0.0136	0.8053 ± 0.0119
	Offline	0.8057 ± 0.0179	0.8199 ± 0.0173	0.8227 ± 0.0104	0.8236 ± 0.0147	0.8107 ± 0.0121	0.8208 ± 0.0101
	Online (No NCE)	0.7984 ± 0.0158	0.8248 ± 0.0059	0.8345 ± 0.0232	0.7877 ± 0.0235	0.8019 ± 0.0104	0.8030 ± 0.0093
	Online	0.8066 ± 0.0167	0.8390 ± 0.0148	0.8515 ± 0.0152	0.7986 ± 0.0162	0.8198 ± 0.0131	0.8289 ± 0.0135
MHEALTH	Offline (No NCE)	0.9897 ± 0.0014	0.9792 ± 0.0014	0.9875 ± 0.0035	0.9886 ± 0.0019	0.9888 ± 0.0017	0.9789 ± 0.0016
	Offline	0.9989 ± 0.0010	0.9987 ± 0.0012	0.9971 ± 0.0042	0.9976 ± 0.0034	0.9981 ± 0.0017	0.9982 ± 0.0016
	Online (No NCE)	0.9300 ± 0.0120	0.9896 ± 0.0006	0.9556 ± 0.0044	0.9765 ± 0.0041	0.9579 ± 0.0021	0.9781 ± 0.0019
	Online	0.9540 ± 0.0011	0.9986 ± 0.0012	0.9672 ± 0.0015	0.9983 ± 0.0012	0.9684 ± 0.0012	0.9984 ± 0.0011

costs. The results of this comparison, presented in Tables 15 and 16, offer insights into the trade-offs between sampling accuracy and computational efficiency. The obvious improvement can be seen through the visualization shown in Fig. 7, where KDE sampling methods can sample more effectively in the feature space compared to Gaussian instance sampling.

Kernel density estimate (KDE), Gaussian kernel, and sampling are expressed as follows:

$$\hat{f}(\mathbf{x}) = \frac{1}{nh^d} \sum_{i=1}^n K\left(\frac{\mathbf{x} - \mathbf{x}_i}{h}\right) \quad (10)$$

$$K(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}u^\top u\right) \quad (11)$$

$$\tilde{Z}_S \sim \hat{f}(\mathbf{x}) \quad (12)$$

where h is the bandwidth parameter, K is the kernel function and d is the dimensionality.

The ablation studies highlight the substantial benefits of adding contrastive loss, Gaussian replay, and considering different sampling strategies in enhancing both the accuracy and stability of the model.

By incorporating these components, the proposed PACL+ framework outperforms baseline models and ensures better retention of previous knowledge during incremental learning. The choice between Gaussian instance sampling and KDE, however, should be made carefully depending on the computational resources available, as KDE provides higher accuracy at the cost of greater memory usage in practical scenarios. As detailed in Table 17, KDE provides slightly better sampling quality but incurs substantial computational and memory overhead. Specifically, during the fitting phase, KDE uses 132.5 % more psutil memory, 240 % more tracemalloc memory, and has an 81 % higher peak memory usage compared to Gaussian sampling, while its sampling throughput is 5.9 % lower. During the sampling phase, KDE shows 5 % higher psutil memory, 171.6 % higher tracemalloc memory, and 81 % higher peak memory, with Gaussian outperforming KDE by 22.2 % in throughput. These findings indicate that although KDE may offer marginal improvements in sample quality, Gaussian replay is substantially more efficient in terms of memory usage and processing speed. This makes Gaussian replay a more practical choice for resource-constrained continual learning systems, where scalability and real-time performance are critical.

Table 10

Ablation study for the impact of InfoNCE loss evaluating the performance for the incremental step on Offline batch learning and online fine-tuning using the ResNet-18 backbone model.

Dataset	Method	ACC (Seen)	F1 (Seen)	ACC (Unseen)	F1 (Unseen)	Overall Accuracy	Overall F1
WISDM	Offline (No NCE)	0.7182 ± 0.0206	0.7076 ± 0.0141	0.8502 ± 0.0222	0.8783 ± 0.0185	0.7476 ± 0.0161	0.7456 ± 0.0128
	Offline	0.7461 ± 0.0080	0.7432 ± 0.0070	0.8837 ± 0.0094	0.8909 ± 0.0118	0.7768 ± 0.0073	0.7760 ± 0.0059
	Online (No NCE)	0.7457 ± 0.0106	0.7410 ± 0.0103	0.8746 ± 0.0241	0.8905 ± 0.0148	0.7744 ± 0.0092	0.7742 ± 0.0096
	Online	0.7463 ± 0.0172	0.7610 ± 0.0144	0.8924 ± 0.0146	0.8877 ± 0.0105	0.7890 ± 0.0135	0.7892 ± 0.0131
REALWORLD	Offline (No NCE)	0.8201 ± 0.0167	0.8331 ± 0.0116	0.8084 ± 0.0287	0.8153 ± 0.0279	0.8166 ± 0.0128	0.8286 ± 0.0095
	Offline	0.8175 ± 0.0108	0.8375 ± 0.0121	0.8458 ± 0.0246	0.8316 ± 0.0219	0.8259 ± 0.0111	0.8360 ± 0.0096
	Online (No NCE)	0.8059 ± 0.0173	0.8085 ± 0.0102	0.8478 ± 0.0214	0.8054 ± 0.0198	0.8153 ± 0.0117	0.8377 ± 0.0105
	Online	0.8050 ± 0.0150	0.8429 ± 0.0078	0.8733 ± 0.0296	0.8066 ± 0.0208	0.8251 ± 0.0103	0.8338 ± 0.0090
MHEALTH	Offline (No NCE)	1.0000 ± 0.0000	1.0000 ± 0.0000	0.9981 ± 0.0015	0.9983 ± 0.0014	0.9991 ± 0.0007	0.9992 ± 0.0007
	Offline	1.0000 ± 0.0000	1.0000 ± 0.0000	0.9984 ± 0.0025	0.9986 ± 0.0022	0.9993 ± 0.0012	0.9993 ± 0.0011
	Online (No NCE)	1.0000 ± 0.0000	0.9999 ± 0.0005	0.9891 ± 0.0013	0.9899 ± 0.0011	0.9896 ± 0.0006	0.9996 ± 0.0006
	Online	1.0000 ± 0.0000	0.9999 ± 0.0003	0.9978 ± 0.0024	0.9982 ± 0.0021	0.9990 ± 0.0011	0.9990 ± 0.0010

Table 11

Ablation study of the proposed PACL+ evaluating the performance for initial subjects using the TinyHAR backbone model.

Dataset	Method	Initial ACC ↑	Initial F1 ↑	Incremental ACC ↑	Incremental F1 ↑	Forgetting ACC ↓	Forgetting F1 ↓
WISDM	No Contrastive	0.7322 ± 0.0128	0.7308 ± 0.0158	0.6173 ± 0.0127	0.6197 ± 0.0114	0.1149	0.1111
	Pairwise Contrastive	0.6379 ± 0.0346	0.6335 ± 0.0366	0.5295 ± 0.0353	0.5149 ± 0.0483	0.1084	0.1186
	No Gaussian Replay	0.7564 ± 0.0132	0.7592 ± 0.0126	0.6252 ± 0.0164	0.6287 ± 0.0143	0.1347	0.1305
	PACL+	0.7564 ± 0.0119	0.7560 ± 0.0128	0.6605 ± 0.0131	0.6635 ± 0.0134	0.0959	0.0925
REALWORLD	No Contrastive	0.8350 ± 0.0105	0.8498 ± 0.0077	0.6055 ± 0.0219	0.6580 ± 0.0178	0.2295	0.1918
	Pairwise Contrastive	0.8400 ± 0.0121	0.8541 ± 0.0106	0.6355 ± 0.0318	0.6297 ± 0.0421	0.2045	0.2244
	No Gaussian Replay	0.8353 ± 0.0122	0.8485 ± 0.0081	0.6235 ± 0.0178	0.6101 ± 0.0130	0.2118	0.2384
	PACL+	0.8353 ± 0.0122	0.8485 ± 0.0081	0.6953 ± 0.0290	0.6781 ± 0.0288	0.1400	0.1704
MHEALTH	No Contrastive	0.9946 ± 0.0027	0.9946 ± 0.0027	0.9728 ± 0.0169	0.9729 ± 0.0168	0.0218	0.0217
	Pairwise Contrastive	0.9969 ± 0.0016	0.9969 ± 0.0017	0.9584 ± 0.0115	0.9585 ± 0.0114	0.0385	0.0384
	No Gaussian Replay	0.9912 ± 0.0035	0.9913 ± 0.0035	0.9208 ± 0.9183	0.9063 ± 0.0149	0.0704	0.0850
	PACL+	0.9912 ± 0.0067	0.9912 ± 0.0069	0.977 ± 0.0141	0.9772 ± 0.0139	0.0142	0.0140

Table 12

Ablation study of the proposed PACL+ evaluating the performance of initial subjects using the ResNet-18 backbone model.

Dataset	Method	Initial ACC ↑	Initial F1 ↑	Incremental ACC ↑	Incremental F1 ↑	Forgetting ACC ↓	Forgetting F1 ↓
WISDM	No Contrastive	0.7461 ± 0.0082	0.7466 ± 0.0077	0.5961 ± 0.0113	0.5941 ± 0.0126	0.1500	0.1525
	Pairwise Contrastive	0.7265 ± 0.0215	0.7264 ± 0.0226	0.5313 ± 0.0208	0.5015 ± 0.0264	0.1952	0.2249
	No Gaussian Replay	0.7547 ± 0.0119	0.7537 ± 0.0109	0.6050 ± 0.0195	0.6189 ± 0.0202	0.1497	0.1348
	PACL+	0.7547 ± 0.0119	0.7537 ± 0.0109	0.6571 ± 0.0208	0.6814 ± 0.0221	0.0976	0.0723
REALWORLD	No Contrastive	0.8410 ± 0.0078	0.8550 ± 0.0078	0.6670 ± 0.0272	0.6963 ± 0.0300	0.1740	0.1587
	Pairwise Contrastive	0.8471 ± 0.0078	0.8601 ± 0.0068	0.6849 ± 0.0626	0.5913 ± 0.0875	0.1622	0.2688
	No Gaussian Replay	0.8461 ± 0.0182	0.8582 ± 0.0166	0.6654 ± 0.0180	0.6685 ± 0.0206	0.1807	0.1897
	PACL+	0.8461 ± 0.0182	0.8582 ± 0.0166	0.7095 ± 0.0338	0.7192 ± 0.0316	0.1366	0.1390
MHEALTH	No Contrastive	0.9969 ± 0.0029	0.9968 ± 0.0030	0.9797 ± 0.0006	0.9797 ± 0.0006	0.0172	0.0171
	Pairwise Contrastive	0.9972 ± 0.0020	0.9971 ± 0.0021	0.9697 ± 0.0006	0.9697 ± 0.0006	0.0275	0.0274
	No Gaussian Replay	0.9952 ± 0.0040	0.9950 ± 0.0042	0.9222 ± 0.0390	0.9296 ± 0.0453	0.0730	0.0654
	PACL+	0.9952 ± 0.0040	0.9950 ± 0.0042	0.9899 ± 0.0007	0.9934 ± 0.0007	0.0053	0.0016

Table 13

Ablation study of the proposed PACL+ evaluating the performance for the incremental step in terms of accuracy and F1 score using the TinyHAR backbone model.

Dataset	Method	ACC (Seen)	F1 (Seen)	ACC (Unseen)	F1 (Unseen)	Overall ACC	Overall F1
WISDM	No Contrastive	0.6925 ± 0.0275	0.6889 ± 0.0225	0.8832 ± 0.0078	0.8820 ± 0.0204	0.7350 ± 0.0226	0.7318 ± 0.0212
	Pairwise Contrastive	0.6791 ± 0.0147	0.6613 ± 0.0165	0.8329 ± 0.0391	0.8554 ± 0.0255	0.7133 ± 0.0171	0.7044 ± 0.0171
	No Gaussian Replay	0.7618 ± 0.0138	0.7645 ± 0.0119	0.8913 ± 0.0110	0.8777 ± 0.0165	0.7906 ± 0.0121	0.7896 ± 0.0113
	PACL+	0.7777 ± 0.0161	0.7584 ± 0.0152	0.9094 ± 0.0093	0.8848 ± 0.0115	0.8170 ± 0.0140	0.8100 ± 0.0132
REALWORLD	No Contrastive	0.8067 ± 0.0156	0.8389 ± 0.0067	0.8533 ± 0.0243	0.8051 ± 0.0252	0.8206 ± 0.0099	0.8305 ± 0.0085
	Pairwise Contrastive	0.8139 ± 0.0151	0.8391 ± 0.0085	0.8421 ± 0.0212	0.8093 ± 0.0235	0.8223 ± 0.0076	0.8316 ± 0.0075
	No Gaussian Replay	0.8066 ± 0.0167	0.839 ± 0.0148	0.8515 ± 0.0152	0.7986 ± 0.0162	0.8198 ± 0.0131	0.8289 ± 0.0135
	PACL+	0.8281 ± 0.0078	0.8316 ± 0.0061	0.8656 ± 0.0111	0.7931 ± 0.0163	0.8321 ± 0.0049	0.8409 ± 0.0046
MHEALTH	No Contrastive	0.9983 ± 0.0022	0.9979 ± 0.0026	0.9969 ± 0.004	0.9976 ± 0.0031	0.9976 ± 0.003	0.7959 ± 0.0028
	Pairwise Contrastive	0.9978 ± 0.0014	0.9354 ± 0.0801	0.5975 ± 0.4879	0.5973 ± 0.4877	0.8095 ± 0.2293	0.7664 ± 0.2826
	No Gaussian Replay	1.0000 ± 0.0000	0.9996 ± 0.0006	0.9568 ± 0.0036	0.9676 ± 0.0032	0.9885 ± 0.0017	0.9986 ± 0.0016
	PACL+	1.0000 ± 0.0000	0.9992 ± 0.0011	0.9965 ± 0.0062	0.9961 ± 0.0059	0.9978 ± 0.0028	0.9976 ± 0.0032

Table 14

Ablation study of the proposed PACL+ evaluating the performance for the incremental step in terms of accuracy and F1 score using the ResNet-18 backbone model.

Dataset	Method	ACC (Seen)	F1 (Seen)	ACC (Unseen)	F1 (Unseen)	Overall ACC	Overall F1
WISDM	No Contrastive	0.6910 ± 0.0153	0.6780 ± 0.0113	0.8939 ± 0.0190	0.8960 ± 0.0152	0.7362 ± 0.0122	0.7264 ± 0.0110
	Pairwise Contrastive	0.6801 ± 0.0152	0.6599 ± 0.0220	0.8052 ± 0.0390	0.8504 ± 0.0342	0.7080 ± 0.0200	0.7023 ± 0.0242
	No Gaussian Replay	0.7593 ± 0.0172	0.7610 ± 0.0144	0.8924 ± 0.0146	0.8877 ± 0.0105	0.7890 ± 0.0135	0.7892 ± 0.0131
	PACL+	0.7756 ± 0.0169	0.7748 ± 0.0155	0.8849 ± 0.0170	0.8996 ± 0.0143	0.8066 ± 0.0140	0.8068 ± 0.0133
REALWORLD	No Contrastive	0.7996 ± 0.0149	0.8381 ± 0.0067	0.8701 ± 0.0119	0.8116 ± 0.0159	0.8202 ± 0.0111	0.8315 ± 0.0089
	Pairwise Contrastive	0.8230 ± 0.0207	0.8352 ± 0.0139	0.7893 ± 0.0342	0.7938 ± 0.0197	0.8134 ± 0.0154	0.8249 ± 0.0131
	No Gaussian Replay	0.805 ± 0.015	0.8429 ± 0.0078	0.8533 ± 0.0296	0.8066 ± 0.0208	0.8151 ± 0.0103	0.8338 ± 0.009
	PACL+	0.8221 ± 0.0228	0.8269 ± 0.0128	0.8756 ± 0.0154	0.7972 ± 0.0195	0.8308 ± 0.0131	0.8195 ± 0.0125
MHEALTH	No Contrastive	1.0000 ± 0.0000	1.0000 ± 0.0000	0.9981 ± 0.003	0.9983 ± 0.0027	0.9991 ± 0.0014	0.9992 ± 0.0013
	Pairwise Contrastive	1.0000 ± 0	0.9997 ± 0.0003	0.9972 ± 0.0018	0.9978 ± 0.0017	0.9987 ± 0.0009	0.9987 ± 0.0008
	No Gaussian Replay	1.0000 ± 0.0000	1.0000 ± 0.0000	0.9975 ± 0.0024	0.9978 ± 0.0021	0.9988 ± 0.0011	0.9989 ± 0.001
	PACL+	1.0000 ± 0.0000	1.0000 ± 0.0000	0.999 ± 0.0013	0.9992 ± 0.0011	0.9996 ± 0.0006	0.9996 ± 0.0006

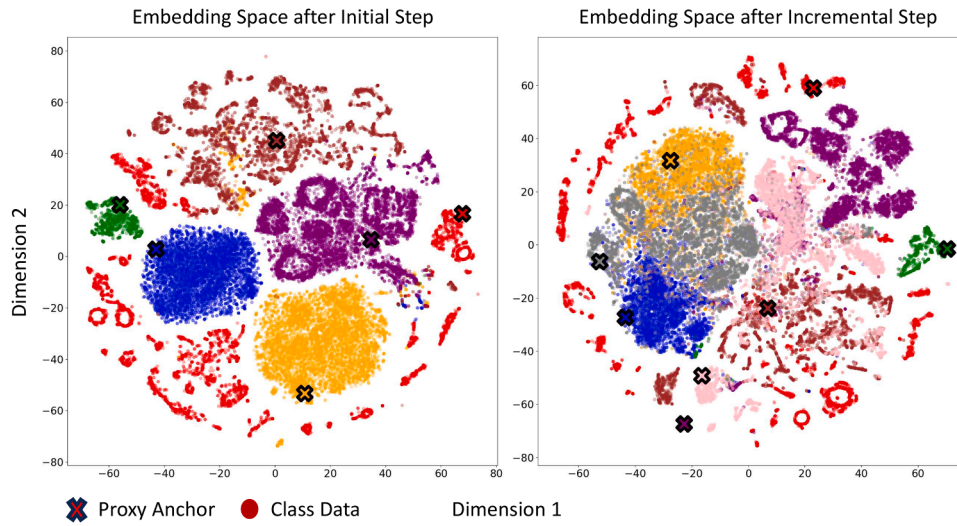


Fig. 6. t-SNE visualization of the embedding space in the REALWORLD dataset (Sztyler & Stuckenschmidt, 2016) after the initial and incremental learning steps. The initially seen classes (left) and the newly seen classes (right) are shown along with their respective proxy anchors. A noticeable shift in the embedding space of all classes highlights the impact of incremental learning, contributing to catastrophic forgetting.

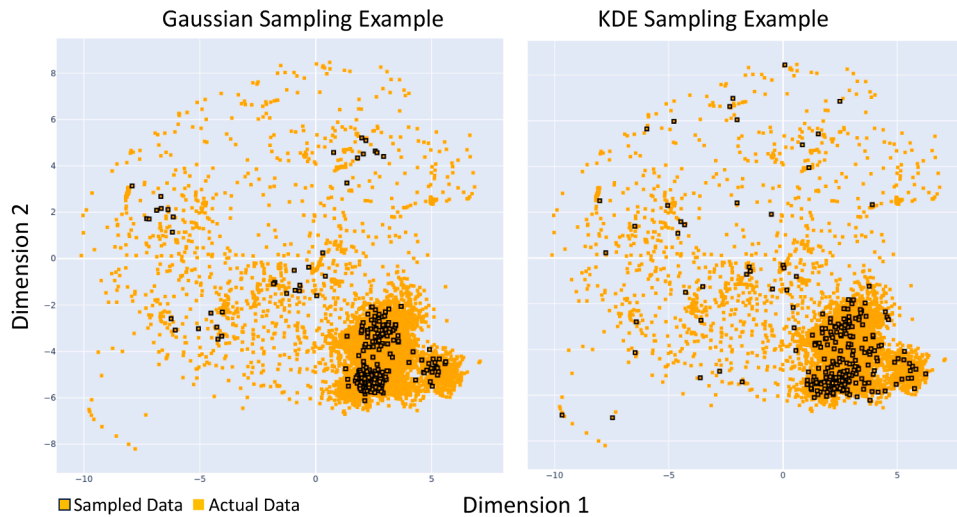


Fig. 7. The difference in embedding space between Gaussian sampling (left) and KDE sampling (right) for the “Climbing Down” class in the REALWORLD dataset (Sztyler & Stuckenschmidt, 2016). Gaussian sampling generates samples concentrated around the cluster center, whereas KDE sampling distributes samples across the entire cluster, capturing a broader representation of the activity. Still, KDE sampling requires substantially higher storage and computational resources than Gaussian sampling, and can thus be considered an upper bound for our approach.

Table 15

Ablation study by comparing the proposed Gaussian Replay with Kernel Density Estimation (KDE) on initial subjects in terms of accuracy and F1 score using the TinyHAR backbone model.

Dataset	Method	Initial ACC $\uparrow$	Initial F1 $\uparrow$	Incremental ACC $\uparrow$	Incremental F1 $\uparrow$	Forgetting ACC $\downarrow$	Forgetting F1 $\downarrow$
WISDM	Gaussian Replay	0.7564 $\pm$ 0.0119	0.7560 $\pm$ 0.0128	0.6605 $\pm$ 0.0131	0.6635 $\pm$ 0.0134	0.0959	0.0925
	KDE Sampling	0.7564 $\pm$ 0.0119	0.7560 $\pm$ 0.0128	0.6888 $\pm$ 0.0135	0.6911 $\pm$ 0.0137	0.0676	0.0649
REALWORLD	Gaussian Replay	0.8353 $\pm$ 0.0122	0.8485 $\pm$ 0.0081	0.6953 $\pm$ 0.0290	0.6781 $\pm$ 0.0288	0.1400	0.1704
	KDE Sampling	0.8353 $\pm$ 0.0122	0.8485 $\pm$ 0.0081	0.7394 $\pm$ 0.0240	0.7650 $\pm$ 0.0183	0.0959	0.0835
MHEALTH	Gaussian Replay	0.9912 $\pm$ 0.0067	0.9912 $\pm$ 0.0069	0.977 $\pm$ 0.0141	0.9772 $\pm$ 0.0139	0.0142	0.0140
	KDE Sampling	0.9912 $\pm$ 0.0067	0.9912 $\pm$ 0.0069	0.991 $\pm$ 0.0006	0.99 $\pm$ 0.0006	0.0002	0.0012

Table 16

Ablation study by comparing the proposed Gaussian Replay with Kernel Density Estimation (KDE) on initial subjects in terms of accuracy and F1 score using the ResNet-18 backbone model.

Dataset	Method	Initial ACC $\uparrow$	Initial F1 $\uparrow$	Incremental ACC $\uparrow$	Incremental F1 $\uparrow$	Forgetting ACC $\downarrow$	Forgetting F1 $\downarrow$
WISDM	Gaussian Replay	0.7547 $\pm$ 0.0119	0.7537 $\pm$ 0.0109	0.6571 $\pm$ 0.0208	0.6814 $\pm$ 0.0221	0.0976	0.0723
	KDE Sampling	0.7547 $\pm$ 0.0119	0.7537 $\pm$ 0.0109	0.6982 $\pm$ 0.0159	0.7001 $\pm$ 0.0150	0.0565	0.0536
REALWORLD	Gaussian Replay	0.8461 $\pm$ 0.0182	0.8582 $\pm$ 0.0166	0.7095 $\pm$ 0.0338	0.7192 $\pm$ 0.0316	0.1366	0.1390
	KDE Sampling	0.8461 $\pm$ 0.0182	0.8582 $\pm$ 0.0166	0.7565 $\pm$ 0.0241	0.7798 $\pm$ 0.0217	0.0896	0.0784
MHEALTH	Gaussian Replay	0.9952 $\pm$ 0.0040	0.9950 $\pm$ 0.0042	0.9899 $\pm$ 0.0007	0.9934 $\pm$ 0.0007	0.0053	0.0016
	KDE Sampling	0.9952 $\pm$ 0.0040	0.9950 $\pm$ 0.0042	0.9942 $\pm$ 0.0006	0.9942 $\pm$ 0.0006	0.0010	0.0008

Table 17

Comparison of memory usage and throughput between Gaussian and KDE methods, showing psutil and tracemalloc metrics for both fitting and sampling phases.

Method	Fitting				Per Sampling			
	Psutil (MB)	Tracemalloc (MB)		#Samples/Sec	Psutil (MB)	Tracemalloc (MB)		#Samples/Sec
		Increase	Peak			Increase	Peak	
Gaussian	64.15	44.68	107.09	4966	19.06	55.65	107.09	68,873
KDE	149.18	140.15	193.79	4672	20.01	151.12	193.79	56,350

4.4. Discussion

Our proposed method has the potential to improve upon existing continual learning approaches for HAR data, particularly in ubiquitous computing systems where processing power is limited. By integrating Proxy Anchor loss with contrastive learning (InfoNCE loss) and Gaussian replay, our approach significantly improves incremental learning performance, striking a balance between high accuracy and minimal forgetting. These advancements make real-world continual learning applications more feasible, especially in scenarios where new activities and subjects must be continuously learned while preserving prior knowledge.

However, our PACL+ has several limitations to be utilized in real-world HAR systems. First, while the experiments in this paper demonstrated our framework successfully mitigates catastrophic forgetting in short-term incremental learning settings, its ability to generalize across multiple incremental steps (N-step learning) remains an open challenge. In practical deployment scenarios, continual learning models must operate within robust and scalable workflows that can handle multiple incremental steps while integrating new classes and subjects seamlessly. One key challenge is ensuring stability across these incremental updates while preserving knowledge from previously seen activities without significant degradation. We plan to extend our experimental design with multiple incremental steps to generalize the proposed method to N incremental steps while efficiently maintaining information about previously learned classes.

Second, real-time adaptation is crucial in dynamic environments where sensor data distributions shift due to variations in user behavior, device placement, or external factors. While our method achieves strong performance on benchmark datasets, its robustness in highly dynamic settings, where sensor data distributions shift over time, needs

further validation. Exploring techniques such as contrastive learning and meta-learning could enhance the adaptability of our model across diverse real-world conditions, ensuring long-term reliability and robustness in continual learning systems.

Lastly, our approach relies on supervised learning with labeled data for continual updates. However, in many practical scenarios, obtaining labeled data for new activities is expensive and impractical. Future work could explore self-supervised or unsupervised learning methods to enable automatic adaptation to unseen activity classes without requiring extensive manual annotation.

5. Conclusion

In this paper, we proposed PACL+, a novel continual learning framework for HAR that integrates Proxy Anchor loss, contrastive learning, and Gaussian replay to mitigate catastrophic forgetting while maintaining high classification accuracy. Our approach effectively addresses key challenges in incremental HAR, including knowledge retention, data imbalance, and adaptation to new activities and users. We evaluated PACL+ on three widely used HAR benchmark datasets, demonstrating superior performance over existing state-of-the-art continual learning methods. Our results show that PACL+ successfully preserves knowledge of previously learned activities for previously learned users, reduces interference from new classes, and improves overall generalization. Detailed ablation studies validated the individual contributions of each component, including contrastive loss and Gaussian replay. Additionally, we presented a comparative analysis of KDE and Gaussian sampling, highlighting the practical efficiency of Gaussian replay for memory-constrained scenarios. The t-SNE visualizations further highlight the model's ability to maintain well-structured feature embeddings,

ensuring robust representation learning in a realistic continual learning scenario.

Despite the significant improvements, the current model size poses challenges for integration into HAR systems with limited memory and computational capacity. As future work, we plan to extend PACL+ to support large-scale continual learning across multiple incremental steps, and explore lightweight architectures and unsupervised strategies to enable real-time, adaptive HAR in dynamic, real-world environments.

CRedit authorship contribution statement

Dhruv Aditya Mittal: Methodology, Software, Validation, Writing-Original draft preparation, Investigation, Data Curation; **Vitor Fortes Rey:** Methodology, Software, Writing-Original draft preparation, Writing-Reviewing and Editing, Visualization, Formal Analysis, Investigation; **Hymalai Bello:** Writing-Reviewing and Editing, Visualization, Formal Analysis; **Paul Lukowicz:** Supervision, Project administration, Funding acquisition; **Sungho Suh:** Conceptualization, Methodology, Formal Analysis, Writing-Reviewing and Editing, Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The research reported in this paper was supported by the BMBF in the project VidGenSense (01IW21003) and Carl-Zeiss Stiftung under the Sustainable Embedded AI project (P2021-02-009). This work was also supported by the Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Ministry of Science and ICT in South Korea (MSIT) (No. RS-2019-II190079, Artificial Intelligence Graduate School Program (Korea University)).

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